



Vehicle positioning systems in tunnel environments: a review

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Abstract

Real-time, accurate, and robust positioning system plays a crucial role in many vehicular applications for automatic driving system and Vehicular Ad-hoc Network (VANET). In the tunnel, the positioning accuracy of Global Navigation Satellite System (GNSS) decreases due to blocked satellite signals. In order to estimate the exact location of vehicles in tunnel environments, many positioning systems have been presented. However, there is a lack of effort in systematically comparing, organizing and analyzing these existing positioning systems, and identifying the strengths and weaknesses of different technologies and applicable scenarios. Therefore, this paper undertakes a thorough investigation into current vehicle localization technologies and methods for tunnel scenarios. The analysis starts with discussing various application scenarios for vehicle positioning system. Then, various vehicle positioning technologies are investigated, the advantages and drawbacks of each technology are illustrated. Thereafter, we discuss some widely used positioning methods in terms of range-based localization method, range-free localization method, multi-sensor fusion localization method, and cooperative positioning (CP) method. Finally, we discuss some challenges faced in vehicle positioning for tunnel environments, and propose some potential research topics for future research work.

Keywords Positioning system · Positioning methods · Tunnel · Vehicle positioning technologies · Vehicular ad-hoc network

Arabic Keywords نظام تحديد المواقع · طرق تحديد المواقع · نفق · تقنيات تحديد مواقع المركبات · الشبكة المؤقتة للمركبات

Introduction

Tunnel environment has a strong confinement, dark light, limited space and other limitations. When a fire accident or traffic accident occurs, if the vehicle and personnel are

not ruttet out in time, it is very easy to occur "second accidents", and often cause far greater harm or injury than the first accident. When encountering sudden accidents in tunnels, if there is no accurate location information, it will affect the rescue and relief, safety and rescue efficiency. Therefore, an efficient, accurate and safe positioning system for the tunnel environment is crucial. The positioning systems are used to locate vehicles and personnels in tunnels, this paper focuses on the vehicle positioning technology and method. Accurate vehicle location information is not only important for preventing traffic accidents, improving rescue efficiency, and easing traffic congestion, but also contributes to the development of automatic driving system and Intelligent Transportation System (ITS) [1–3].

The vehicle positioning in tunnels is different from outside tunnel. Outside tunnel, the Global Navigation Satellite System (GNSS)-based positioning strategy usually used to locate a vehicle. However, in the tunnel, due to the obstruction of the tunnel walls, the satellite signal cannot be fully covered [4], therefore the traditional satellite-based positioning method cannot accurately realize the vehicle localization, and the localization accuracy cannot meet the requirements of the applications. In order to tackle these issues, various

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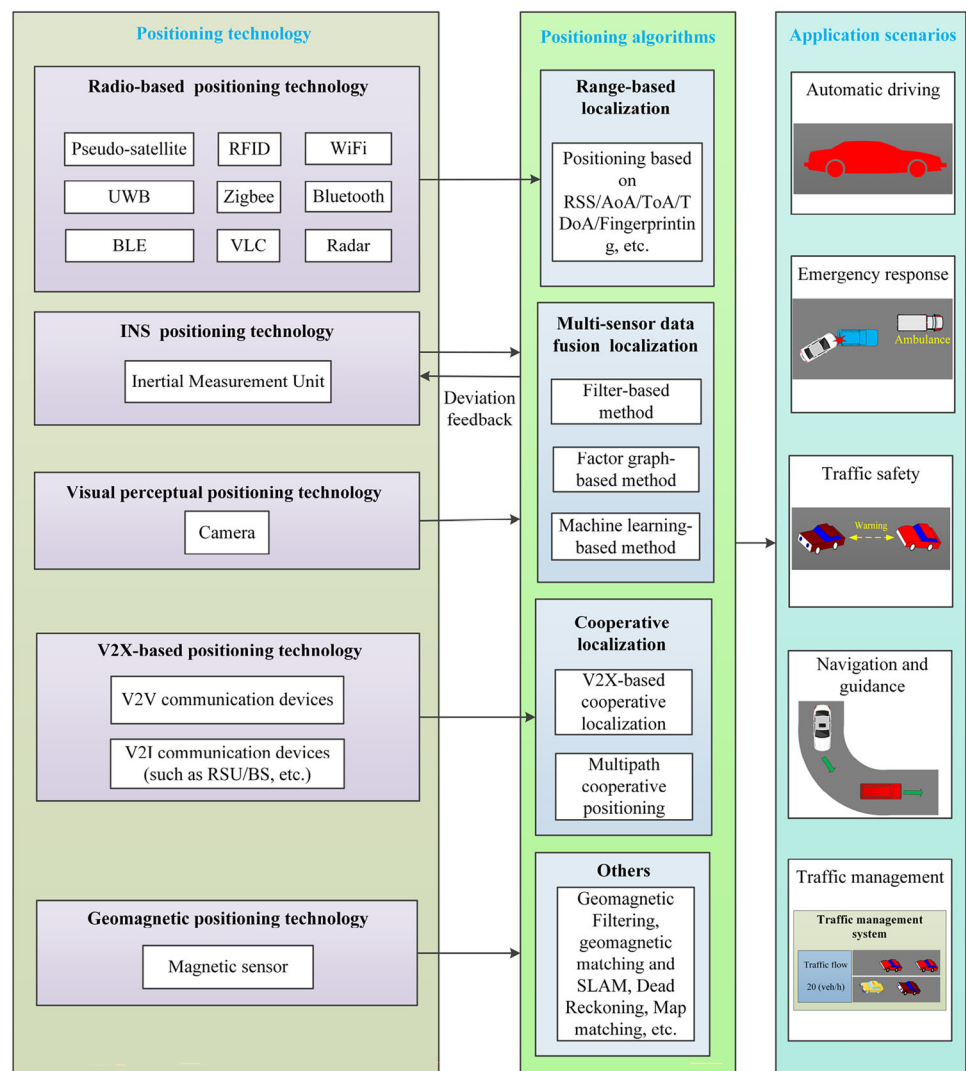
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Fig. 1 The overall architecture of the vehicle positioning system for the tunnel environment



vehicle location solutions using various technologies have been proposed. Figure 1 illustrates the vehicle positioning architecture in the tunnel environment. According to the requirements for positioning accuracy and real-time performance in different application scenarios, the appropriate positioning technology is selected, and then advanced positioning algorithms are used to realize more accurate vehicle positioning to meet the actual needs.

The state-of-the-art vehicle positioning systems encompass a variety of technologies, such as radio-based on localization, Inertial Navigation System (INS), visual perception-based positioning, Vehicle-to-Everything (V2X)-based positioning, and geomagnetic field-based positioning, etc. Various types of sensors are employed in these positioning systems to collect information, including inertial sensors, pseudolite receivers, wireless receivers, photodiodes, radar, Light Detection and Ranging (LiDAR), magnetometers, and cameras, etc. Radio-based localization methods, such as those using pseudo-satellite [5], and other radio signals,

provide wide-area coverage and are widely used due to their high availability and ease of integration. However, their performance degrade in tunnels due to signal obstruction and multipath effects. INS offer continuous positioning by utilizing accelerometers and gyroscopes to track vehicle movement, making them invaluable in environments where radio signals are weak or unavailable. Nevertheless, INS suffer from cumulative drift over time, requiring periodic correction from other sensors. Visual perception-based positioning leverages cameras and computer vision techniques to interpret the vehicle's surroundings, enabling precise vehicle localization in visually rich environments. This method is suitable for scenarios with an abundance of visual landmarks, but may be affected by poor lighting conditions, adverse weather, occlusions, object shapes and textures. The tunnel is dimly lit, which will make the visual localization accuracy will be reduced. Vehicle positioning based on V2X is a promising positioning technology. The information sharing can be realized through V2X com-

Table 1 The review papers on vehicle localization

References	Year	Scenario	Focus
[9]	2017	Indoor	Different positioning approaches
[10]	2021	Indoor	RFID-based vehicle localization
[11]	2021	VANET	Various vehicle positioning techniques
[1]	2018	VANET	Localization prediction approaches
[12]	2021	Highway	Various localization methods
[13]	2022	ITS	State-of-the-art (SoA) vehicle positioning systems
[14]	2019	Smart cities	SoA vehicle positioning techniques
[15]	2018	Autonomous driving	SoA localization techniques
[16]	2023	Autonomous driving	SoA localization approaches
[17]	2023	Autonomous driving	Simultaneous localization and mapping (SLAM)
[2]	2022	Autonomous driving	Visual SLAM methods
[18]	2020	Autonomous driving	Stereo Visual SLAM
[19]	2017	Autonomous driving	SLAM
[20]	2023	Autonomous driving	Map-Based localization techniques
[21]	2022	Autonomous driving	SoA self-localization techniques
[22]	2020	Autonomous driving	Three-dimensional (3D) LiDAR localization
[23]	2020	Autonomous driving	Deep learning (DL) sensor fusion
[24]	2024	–	Visible light positioning method
[25]	2023	–	SoA localization techniques and methods
[26]	2017	–	Vehicle relative positioning
[27]	2023	–	State estimation algorithms
[8]	2021	–	V2X-based positioning methods

munication technology, which enables vehicles to exchange information with each other vehicle and with infrastructure elements such as Base Station (BS) and Road Side Unit (RSU) [6, 7]. Wireless communication technology provides an information interaction platform for vehicle cooperative positioning (CP), where V2X communications can be realized by using Long Term Evolution-Vehicle (LTE-V) communication or Dedicated Short Range Communication (DSRC) [8], etc. V2X communication-based positioning can enhance positioning accuracy and enable cooperative localization in connected vehicle scenarios. Eventhrough there are many advantages to CP, there are also some disadvantages. V2X-based cooperative positioning system has high cost and complexity. It usually requires communication between vehicles or between vehicles and BSs, and the performance of the localization system may be influenced if the communication network is unstable or there is some interference. Geo-magnetic field-based positioning uses the Earth's magnetic field variations to determine location, providing a complementary method in environments where other systems may falter. However, this technique requires detailed geomagnetic maps and can be influenced by local magnetic disturbances. Each of these positioning systems brings unique strengths and weaknesses, combining multiple localization methods or employing sensor fusion can further enhance the accuracy

and robustness of vehicle localization in tunnels, improving safety and navigation for vehicles in enclosed environments.

The accuracy of vehicle positioning depends not only on the positioning technology used but also closely on the specific positioning methods employed. Many studies have been conducted to study high precision vehicle positioning methods for the tunnel scenarios. The common vehicle positioning methods encompass range-based methods [e.g., Time of Arrival (ToA), Time Difference of Arrival (TDoA), Angle of Arrival (AoA), Received Signal Strength (RSS)], range-free methods (e.g., fingerprinting), multi-sensor fusion approaches, and cooperative positioning strategies. The range-based localization methods can offer high accuracy but are susceptible to multipath effects and signal attenuation. Range-free methods, like fingerprinting, are less affected by multipath but may have lower accuracy. Multi-sensor fusion methods can improve accuracy by integrating data from multiple sensors [such as Inertial Measurement Unit (IMU), LiDAR, and cameras, etc.], but they are complex to implement and calibrate. Cooperative positioning methods can enhance accuracy by sharing information between vehicles or infrastructure, but they depend on reliable communication, which may not always be available in tunnel environments. This paper reviews these methods and discusses their advantages and disadvantages, aiming to

provide insights into improving vehicle positioning in tunnels.

There have been many reviews on vehicle localization technologies as shown in Table 1. It can be seen that existing reviews are focused on indoor scenario, outdoor scenario, Vehicular Ad-hoc Network (VANET) scenario, ITS, and autonomous driving, etc. However few reviews focus on vehicle localization in tunnel environments. In this paper, we will present a comprehensive overview of vehicle positioning systems in tunnel environments. To do this comprehensively, we provide a brief summary on the applications of vehicle localization in tunnels, such as automatic driving, emergency response, traffic safety, navigation and guidance, and traffic management, etc. Furthermore, we survey several positioning technologies in tunnels. In addition, the widely used positioning methods are summarized in this paper. At last, we discuss some challenges and future directions for vehicle localization in tunnels. The main contributions of this paper are summarized as follows:

- (1) This survey provides a detail survey of various existing vehicle positioning techniques for tunnel environments in the literature including positioning principles, the advantages and disadvantages of various positioning techniques, etc.
- (2) This survey provides an in-depth exploration of vehicle localization methods utilized in tunnel environments, including the range-based positioning method, the range-free positioning method, the multi-sensor fusion positioning method, CP method.
- (3) This survey presents some challenges of vehicle positioning technologies in tunnel environments. Moreover, the future perspectives of vehicle positioning system are discussed.
- (4) The existing and potential applications of tunnel vehicle localization technologies are presented, such as automatic driving, emergency response, traffic safety, navigation and guidance, and traffic management, etc.

The structure of this paper is organized as follows: in Sect. 2, we provide a concise discussion on some applications of vehicle localization in tunnel environments. In Sect. 3, we give the detailed description of different vehicle positioning technologies in the tunnel environment. We compare the advantages and disadvantages of various positioning technologies. In Sect. 4, different vehicle positioning methods in tunnel environments are described, such as the range-based localization methods (i.e., ToA, TDoA, AoA, RSS, etc.), range-free localization methods, the multi-sensor fusion localization methods (i.e., the filter-based method, Factor graph (FG)-based method, machine learning-based method), and CP methods, etc. Then, some challenges of vehicle localization

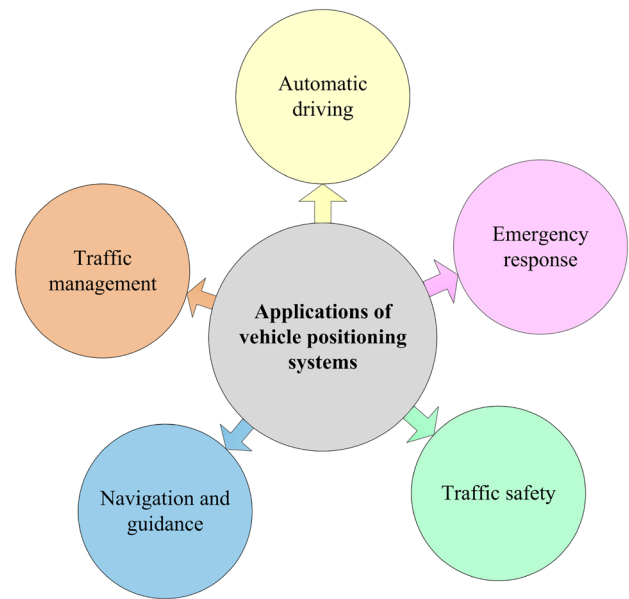


Fig. 2 The applications of vehicle positioning system

technology in tunnel environments are provided in Sect. 5. Then, we provide some potential research topics for future work in Sect. 6. Lastly, some concluding remarks are provided in Sect. 7.

Applications of vehicle localization in the tunnel

Vehicle positioning in tunnels is essential for various applications, primarily focused on traffic safety enforcement, emergency response coordination, navigation and guidance systems, traffic management optimization, operational efficiency enhancement, and overall transportation efficiency improvement, etc. The applications of vehicle positioning are shown in Fig. 2. In the following, we present some key applications of vehicle positioning.

Automatic driving

Vehicle location information is a very important component in autonomous driving systems [28], which can be used to realize autonomous driving localization and navigation, obstacle object monitoring and avoidance, fleet collaboration, autonomous driving decision making, and updating of autonomous driving vehicle maps [29]. Research on autonomous driving is currently underway, with efforts focused on enhancing road transport efficiency, mitigating traffic accidents, and enhancing overall traffic safety [30].

Determining the position of a vehicle within a lane, known as lane-level positioning accuracy, is crucial for applications

such as lane-keeping assistance and lane-level navigation in autonomous driving. The typical requirement for “where in lane” accuracy is around 10–20 cm or better, depending on the specific application and level of autonomy required.

Emergency response

The location of emergency facilities is a key component in emergency response activities [31]. Accurate vehicle positioning allows emergency services to respond quickly to incidents within the tunnel, such as accidents, fires, or medical emergencies. In emergency response, vehicle position sensors greatly impact the safety and efficiency of emergency response operations, and help minimize response times. Accurate vehicle location information facilitates the emergency response teams to reach the accident site as soon as possible. In case of emergencies like tunnel fires or toxic gas leaks, vehicle positioning helps guide tunnel users to safe exit routes and helps rescue teams locate stranded vehicles. This will allow the evacuation of the accident site to be completed as quickly as possible.

In tunnel scenarios, the requirements for vehicle positioning accuracy in emergency response are usually more stringent. Different countries and regions have different requirements for vehicle positioning accuracy. Generally speaking, for tunnel scenarios, vehicle positioning accuracy is required to be within a few meters, or even higher.

Traffic safety

Vehicle location technology plays a crucial role in enhancing traffic safety by providing real-time information about vehicle positions, speeds, and behaviors. These technologies are utilized in traffic safety to prevent accidents and optimize traffic flow [32].

Vehicle location information can be exchanged between vehicles in real time in the VANET. Collision avoidance systems use vehicle location data to detect potential collisions and warn drivers to take evasive action. These systems often incorporate various positioning technology to track the positions of nearby vehicles and calculate the risk of a collision based on their relative speeds and trajectories. If a collision risk is detected, the system can alert the driver through visual or auditory warnings, or even autonomously apply brakes or steer the vehicle to avoid the collision and improving traffic safety.

The accuracy requirements for an ITS can be classified into three categories: which road, which lane, and where in lane. The typical requirement for “which road” is 5 ms or better, which can be easily achieved using current positioning and navigation devices. This level is widely adopted in road planning and congestion relief. The requirement for “which lane”, the standards of navigation performance must be less

than 1.5 ms. This level is capable of not only describing a vehicle’s position on a particular road but also identifying its lane [33]. The requirement for “where in lane” accuracy typically involves determining the exact lateral position of a vehicle within a lane. This level of accuracy is often specified in centimeters. There are several high-precision positioning technologies, such as camera, radar, LiDAR, which can achieve centimeters-level accuracies in favorable environments [8]. For critical ITS safety applications such as forward collision warning, intersection collision warning, lane change warning, and blind spot warning, a positioning accuracy of 0.5–1 m is required [28, 30].

Navigation and guidance

Vehicle positioning technology can provide real-time navigation and wayfinding services to drivers, helping them navigate through complex tunnel networks and reach their destinations efficiently. Vehicle location information can also be used for wrong-way driver warning. Moreover, vehicle positioning data can be used in conjunction with surveillance cameras to track and monitor the movement of vehicles and individuals within the tunnel, enhancing traffic safety.

In navigation and guidance systems, the specific requirements for vehicle positioning accuracy typically depend on the system design and application scenarios. Generally, navigation and guidance systems have high requirements for vehicle positioning accuracy to ensure that vehicles can accurately track routes and execute navigation instructions. Specific requirements may vary depending on the system and application, but generally require accuracy within a few meters. For example, in vehicle autonomous driving systems, navigation and guidance systems typically require vehicle positioning accuracy of around 10–20 cm or even less than 10 cm to ensure that vehicles can accurately identify lanes and road signs, and achieve precise lane-keeping and turning operations. Additionally, in the aviation and maritime fields, navigation and guidance systems have very high requirements for positioning accuracy, typically requiring accuracy within a few meters, to ensure that aircraft or vessels can accurately follow predefined routes and paths. In summary, navigation and guidance systems have high requirements for vehicle positioning accuracy, typically requiring accuracy within a few meters, to ensure the accuracy and reliability of the system.

Traffic management

Real-time vehicle positioning data can be used to monitor and manage traffic flow within the tunnel. Traffic signals, variable speed limits, and lane management systems can be adjusted based on current traffic conditions. In addition, vehicle positioning data can be used to detect and respond to

incidents such as accidents, breakdowns, or debris in the tunnel, reducing congestion and improving overall traffic efficiency. Therefore, accurate vehicle location is important for traffic management and helps enable traffic flow optimization and incident detection [34]. In addition, knowing the precise location of vehicles within the tunnel can help tunnel operators monitor and manage assets like ventilation systems, lighting, and surveillance cameras more effectively. Moreover, vehicle positioning data can be used to predict maintenance needs by tracking the wear and tear of tunnel infrastructure components based on vehicle usage. Vehicle positioning data can be analyzed to conduct traffic studies, assess usage patterns, and plan for future tunnel expansions or improvements. Furthermore, this data can be utilized to evaluate the tunnel's performance concerning traffic flow, safety, and efficiency.

As mentioned above, vehicle positioning in tunnels plays a critical role in enhancing safety, improving traffic management, optimizing operations, and perfecting autonomous driving system. Depending on different application scenarios, vehicle positioning systems can be applied in a variety of fields, such as VANET scenarios, autonomous driving scenarios, intelligent traffic management, emergency rescue and public safety, as well as shared mobility services, etc. Different scenarios have different requirements for positioning accuracy, to be effective in these applications, ultra-high vehicle positioning accuracy is required.

Positioning technologies

Achieving high precision in vehicle positioning, especially in complex environments like tunnels, is a challenging task. To address this challenge, several localization techniques for vehicle positioning in tunnels where GNSS signals are not available have been proposed and studied extensively in the literature. These techniques can be broadly categorized into several types: radio-based positioning technology (such as pseudo-satellite, radio frequency devices, Visible Light Communication (VLC), Radio detection and ranging (Radar), etc.) [35–37], INS [38], visual perceptual localization [39], digital map [40], geomagnetic positioning [41], V2X-based positioning technology [42], etc. Each of these localization techniques has their own advantages and disadvantages. Figure 3 illustrates vehicle positioning inside tunnels. In tunnels, various devices can be installed, such as Radio Frequency Identification (RFID), Ultra-wideband (UWB), WiFi, Zigbee, Bluetooth, Bluetooth Low Energy (BLE), VLC transmitter, BS, and RSU, etc. Vehicles are equipped with corresponding receiving devices, such as VLC receiver, and other wireless receivers. These devices receive signals from the devices installed on tunnel walls, the ground, and other surrounding vehicles. Based on the received sig-

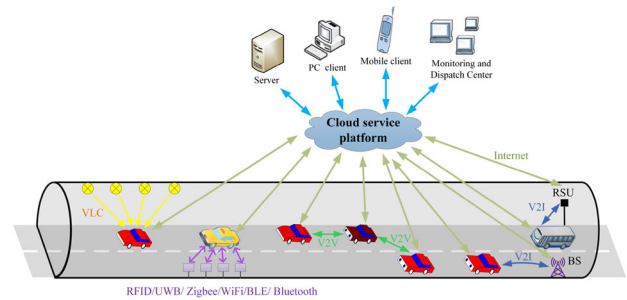


Fig. 3 The vehicle positioning in the tunnel environment

nals, localization algorithms are used to estimate the location of the target vehicle. Certainly, on-board sensors such as cameras, radar, LiDAR, IMU, and ultrasonic sensors can be installed on the vehicle to estimate vehicle position using the information they provide. The estimated position information can be transmitted to the cloud service platform, where users can monitor vehicle locations via computers or mobile devices. The obtained location data can also be forwarded to traffic management authorities, facilitating efficient traffic management and congestion relief to improve the efficiency of road access.

Radio-based positioning techniques

The radio-based localization technology usually estimates a current position of the target vehicle including pseudo-satellite technology, radio frequency technology (i.e., RFID, UWB, WiFi, Zigbee, Bluetooth, BLE, Wireless Sensor Networks (WSN), etc.), VLC, Radar, etc.

Pseudo-satellites-based technology

Because Global Positioning System (GPS) is usually not available for a long time in a tunnel, the GPS-based localization technologies have large positioning errors, and are not suitable for tunnel environment. To overcome the satellite signal blockage in the tunnel, the localization system based on pseudo-satellite technology was proposed. The positioning techniques using pseudo-satellites is a traditional technique in which a transmitter placed on the ground transmits the same message as the GNSS signal, and when the user enters the tunnel space, user receives the GPS/BDS signal, which is then solved for positioning to locate the target in the tunnel. In [43], authors proposed a multi-source fusion localization method based on pseudo-satellite assistance to realize the positioning in a tunnel-like scenario (i.e., indoor long and narrow scenario). The proposed localization method based on pseudo-satellite is suitable for tunnel scenarios. In [5], a pseudo satellite positioning system was designed to estimate vehicle position in the closed environment such as

tunnels. The proposed positioning system used pseudo satellite to broadcast the position of the host station GPS signal. The trilateral positioning algorithm was applied to obtain vehicle location. In [28], a synchronized pseudolite system was proposed to achieve low-cost and high-accuracy vehicle positioning in tunnels. Additionally, the integration mechanization of the vehicle positioning system with a digital elevation model, laser ranging, and an IMU was implemented to enhance the accuracy and stability of three-dimensional vehicle positioning.

Radio frequency-based technology

Positioning technologies based on radio frequency technology includes RFID, UWB, WiFi, Zigbee, Bluetooth, BLE, WSN to determine the position of vehicles within a confined or localized area.

In [44, 45], authors used the RFID-based localization method to estimate the vehicle position, results showed that this positioning method enables accurate estimation of vehicle position and has low-cost. In [46], a multi-sensor fusion strategy was proposed to estimate vehicle position, which include some low-cost sensors, such as RFID, in-vehicle sensors (i.e., wheel speed sensors and electronic compasses) and MEMS-based inertial sensors. In the paper, the preliminary localization algorithm based on RFID and the interactive multiple model (IMM)-strong tracking Extended Kalman Filter (EKF) (STEKF) algorithm were developed to improve the positioning accuracy. In the proposed RFID-based preliminary positioning strategy, the least mean square federated filter was utilized to achieve preliminary localization of vehicle. The results validate the effectiveness of this positioning strategy. The IMM-STEKF-based global fusion algorithm was applied to fuse the data from the in-vehicle sensors and preliminary localization results. In [47], authors proposed a Ultrahigh frequency (UHF) RFID-based vehicle positioning method for the shadow areas with weak or even no GPS signals, for example, tunnels, urban canyons, lower layers of multi-layer bridges, mine, underground parking garage, etc. In [48], authors proposed an RFID-based hybrid vehicle positioning method for GPS-less scenarios, where readers were installed on the target vehicle and passive RFID tags were deployed on the roads. In [49], authors proposed an active RFID localization technology to obtain vehicle position. The proposed localization scheme is also suitable for the scenarios where GNSS signals are not available like underground parking garage or tunnels. In [50], authors introduced a fine-grained backscatter localization strategy using the COTS RFID products with detected phase. So as to improve the positioning accuracy, many scholars have fused multiple positioning techniques to realize vehicle positioning. In [51, 52], the strategy fusing in-vehicle sensors and RFID information was proposed for vehicle localization in

the tunnel. The proposed positioning strategy includes the following steps, namely, calculating the distances between the reader and the RFID tags, and estimating the vehicle location using the global fusion algorithm. The RFID-based localization system requires LoS between tag and reader to ensure high accuracy.

In [53], the authors presented a method to reduce the impact of NLoS transmissions and used UWB positioning technology to achieve vehicle localization. In [54], authors proposed the vehicle localization system and accident detection algorithm for the tunnel environment based on UWB positioning technology, which realize real-time vehicle tracking and localization, and ensure the safety of automatic driving in tunnel. In [55], authors used UWB technology to estimate vehicle positions. In [56], authors used the downlink TDoA (DL-TDOA) based on UWB to acquire precise vehicle locations in the tunnel. The proposed DL-TDoA inherits UWB technology to measure the TDoA information between four different location base stations and the location tag, then calculate the distance differences between four groups of location BSs and the location tag.

In addition, some people use WiFi and BLE for vehicle localization. In [57], the authors used WiFi-based positioning technology to acquire vehicle locations in the underground mine tunnels. In [35], a WiFi fingerprinting localization method was applied to estimate vehicle location in the tunnel. In addition, BLE was also used to locate the vehicle. In [36], authors proposed a positioning strategy based on BLE technology.

The emergence of WSN technology, which can cover a range of several hundred meters, has further expanded the coverage area and application range of short-range wireless communications. In [58], authors proposed an angle offset-assisted positioning (AOAP) system to improve the localization accuracy of railway vehicles in the tunnel. The proposed localization strategy includes a WSN, where a target node is installed the train, and some anchor nodes are deployed along the railway tracks. The target location can be initially calculated using the data from anchor nodes. Then, considering the impact of angle offset error on the positioning precision, the target position is updated. Results showed that the proposed positioning scheme has better performance than the Least Square (LS) method when not accounting for angle offset. In [59], based on the wireless multimedia sensor networks installed in a coal mine, authors proposed a miner lamp video CP method to calculate the miner position in underground tunnels.

VLC

VLC technology gives a new approach to track vehicles in the tunnel scenarios. One of the advantages is that lighting sources can be installed as a part of the establishment of

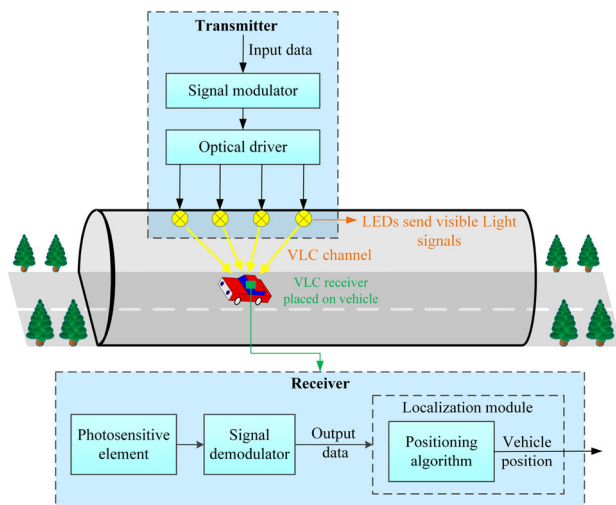


Fig. 4 An example for the general structure of VLC positioning system based on four LEDs

tunnels and does not need to install additional devices into the tunnel [37]. Figure 4 illustrates an example for the general structure of VLC positioning system. VLC positioning system mainly consists of a VLC receiver and transmitter, where VLC transmitter modulates the intensity of an optical source (such as LED) operating in visible range of the electromagnetic spectrum, and VLC receiver extracts the data from the modulated light beam based on a photosensitive element and estimates the vehicle position [60]. The photosensitive element is usually a positive intrinsic-negative photodiode or an image sensor such as camera [61], it can be used to capture visible light signals. Then the visible light data from photosensitive element is processed by the signal demodulator. The output data from signal demodulator is transmitted to the localization module for target location. Based on the obtained output data, various localization algorithm can be used to acquire the target location, such as the trilateration method, multilateration method, vision-aided positioning method, etc. In the VLC positioning system, the VLC transmitter and receiver are connected by the VLC channel. For the VLC system, there is a mandatory condition that Line-of-Sight (LoS) components are presence between the lights and the receiver [62].

In [37], the authors provided a localization strategy for tunnel environment, the system uses the VLC technology and Cell-ID positioning method. In [63], a new 3D VLC-based positioning system was proposed for underground mining environment, which is a tunnel-like environment. The proposed localization system can locate miners by installing a minimum of three LEDs in each sector of the mine and placing a photodetector on the roof of the miner. The signal received by the photodetector is a composite of three LED signals, and an appropriate algorithm is applied to distinguish the signals received by the photodetector. Subsequently, the

power of each LED is calculated. Ultimately, using the estimated power of the received signal, the location of miner is determined through the trilateration localization method. The proposed positioning approach can also be used to locate vehicles and other objects in the tunnel. In [64], authors proposed a VLC system to estimate the location of personnel and machinery in underground mining tunnels. In [65], a reliable visible light localization algorithm was suggested to locate personnel in the underground mine tunnel. The proposed positioning method can also be adopted to locate vehicles in the tunnel. In [66], authors proposed a positioning scheme using LED technology. The proposed positioning method can estimate the target location through the color temperature and frequency ratio using LED with three color temperatures. In [67], VLC technology was used to calculate the position of miner inside underground mine scenario. The proposed positioning scheme can also be utilized to acquire the vehicle location. In [68], a novel vehicle localization scheme based on Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) VLC was presented to locate vehicle in the tunnel. In the proposed localization scheme, the camera sensor was adopted to extract the position data of LED light sources from the tunnel infrastructure and vehicle tail lamp. The vehicle location was estimated based on three LED signals detected by the camera sensor. The VLC using light devices has been considered a promising strategy for vehicle localization in the tunnel. However, currently, vehicle positioning technology based on VLC is still in the developmental stage and has not yet reached maturity or widespread application.

Radar

Automotive radar includes LiDAR [69], Millimeter wave (mmWave) radar [70], ultrasonic radar. All of these radars can be used for vehicle positioning, with LiDAR being the most widely used. LiDAR performs ranging by measuring the time or phase difference between round trips of lasers. LiDAR has extremely high angular and distance resolution, and can generate a large number of high-precision laser point clouds. Based on the distance information of each laser measurement point and its corresponding pulse LoS information, the coordinates of the 3D point cloud in the LiDAR coordinate system can be obtained.

Figure 5 presents a common framework of LiDAR-based positioning system. The LiDAR system emits laser pulses towards the surrounding environment. These laser pulses are typically in the form of short, high-intensity light bursts. When the laser pulses hit objects in the environment (such as other vehicles, buildings, trees, etc.), they get reflected back to the LiDAR sensor. The LiDAR sensor measures the time it takes for the laser pulses to travel to the object and back. This Time of Flight (ToF) information is used to calculate the distance to the objects based on the speed of

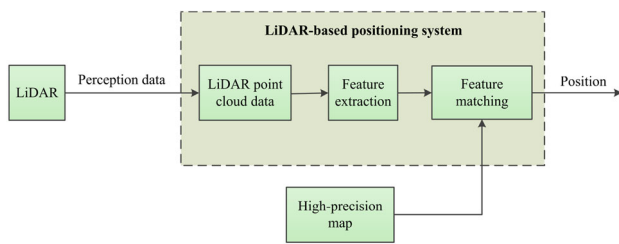


Fig. 5 The framework of LiDAR-based positioning system

light. The LiDAR system rotates or uses multiple lasers to cover a 360° field of view around the vehicle. The angle at which the laser pulses are emitted and received is recorded. Using the distance and angle measurements, the LiDAR system generates a point cloud, which is a 3D representation of the environment. Each point in the cloud represents a reflection point from an object in the environment. Using feature extraction algorithms, representative feature points are extracted from the point cloud data. These feature points can be corners, edge points, or surface points, among others. Extracting these features simplifies the data while retaining critical environmental information. By combining the extracted feature points, a detailed 3D environment map is generated. These maps accurately reflect the geometric structure of the surrounding environment and are used for subsequent vehicle positioning and navigation. By comparing the features of the currently collected point cloud data with those in the pre-generated environment map, the vehicle's position can be determined. Common feature matching methods include nearest neighbor search algorithm, iterative closest point algorithm, and random sample consensus algorithm. Through feature matching, the transformation matrix between the current point cloud and the corresponding feature points in the map is calculated. Based on this transformation matrix, the vehicle's position and orientation in the environment can be estimated.

In [71], authors proposed an infrastructure-free mapping and positioning strategy for getting the velocity and location of the railed vehicle in the tunnel scenario. The proposed localization framework only used a LiDAR sensor. To improve the accuracy and reliability of the vehicle positioning, LiDAR data is often integrated with data from other sensors such as cameras, GPS, and IMUs, etc. In [72], a fusion positioning approach based on roadside LiDAR and IMU was designed to overcome positioning problems in the tunnel environments. In addition, the LiDAR and IMU fusion method was proposed to calculate the vehicle location. Results showed that the average localization error of the proposed localization method is sub-meter level. In [40], 3D LiDAR was used to estimate vehicle location. The fire hydrants and evacuation guide lights installed at the tunnel walls were employed to obtain the longitudinal location.

Different objects have different reflectivity, using LiDAR reflectivity information can distinguish between different objects. In addition, in order to generate a synthetic reflectivity map, authors fused the road surface reflectivity map and the reflectivity information of tunnel walls. After the synthetic reflectivity map was established, the vehicle location can be obtained by single correlation matching with a local map generated from the current LiDAR information. In [73], authors used a LiDAR and a peculiar movement pattern to locate a robot in tunnel-like environment, for example, mines, tunnels or underground pipes. In [69], a LiDAR-based map-aided positioning method was suggested to locate the vehicle in the highway tunnels. In the paper, 3D LiDAR was utilized to extract the tunnel facility points, and vehicle location can be estimated using the EKF algorithm.

Millimeter wave radar technology is a wireless sensing technology with a wide range of applications, which can be used for unmanned and assisted driving of automobiles, localization and so on. The mmWave radar detects the presence, distance, speed and orientation of a vehicle by emitting electromagnetic waves and detecting the echoes. The use of millimeter wave radar for vehicle location has broad prospects. In [74], authors used mmWave radar to locate and map for the scene of fire and smoke in the tunnel. In the paper, authors analyzed the effects of multi-path scattering on SLAM, and proposed a radar azimuth scattering angle signature and two radar false detection revising methods to promote the accuracy of pose estimation.

Some scholars use ultrasonic sensors for vehicle localization. In [75], authors proposed an autonomous localization approach for local region vehicles in coal mine tunnels using ultrasonic and visual sensors. In the proposed positioning approach, the on-board vision sensors can be exploited to recognize barcode labels attached in pairs to tunnel walls. Ultrasonic sensors can be employed to measure the distance from the center point of the vehicle to the tunnel walls. After determining the spatial geometric relationship between the center point of vehicle and the barcode labels, the vehicle position can be calculated. In [76], a joint electromagnetic wave and ultrasonic positioning strategy was proposed.

INS

The INS includes two parts, namely the IMU and the computational unit. The IMU includes accelerometers, gyroscopes and/or magnetometers [77]. The IMU is used to measure angular velocity and acceleration information of the navigation platform relative to the inertial coordinate frame of reference. The computational unit combines the information about the vehicle's acceleration and angular velocity to calculate the velocity, attitude, and location of the vehicle. Dead Reckoning (DR) algorithm is widely adopted to calculate the target positions in INS. The DR method relies on the accel-

eration and angular velocity information obtained from the inertial sensors to recursively derive the current location and heading based on the position and heading calculated at the previous moment.

Since INS suffers from cumulative errors, INS is usually combined with other techniques to achieve highly accurate vehicle positioning. In [38], a novel localization strategy combining UWB and the low-cost microelectromechanical system (MEMS) INS was proposed to track the vehicles in the tunnel environment. The proposed positioning strategy includes a novel FG-based positioning algorithm, an residual weighting (Rwgh)-based NLoS error-mitigation method and an EKF-based loosely coupled UWB/INS integration method. In this strategy, a two-step NLoS error reduction approach was used, specifically, the FG positioning method was employed to calculate vehicle location based on UWB information, which can preliminary mitigate the NLoS error of the radio signal; then a Rwgh algorithm was used to further reduce the NLoS error. After reprocessing by Rwgh-based algorithm, the vehicle location obtained by UWB information can be approximately seen as LoS. Finally, a loosely coupled UWB/INS integration approach was employed to obtain the final vehicle location. In [78], authors improved positioning accuracy by fusing the IMU and UWB data. In [79], based on the GNSS/INS/odometer combination positioning, authors proposed a lane-constrained positioning method that integrates odometer mileage calculations and laser radar lateral distance measurements. In the proposed positioning system, the positioning points calculated from odometer mileage and lateral distance were fused with the INS/odometer combination positioning.

Visual perceptual localization techniques

Visual localization is a technique for determining the location of a target by visually perceiving and analyzing environmental information. Figure 6 presents the framework of the vision-based positioning system. The basic principle of visual positioning is as follows: first, the onboard camera is calibrated to determine the camera's internal parameters (such as focal length, principal point offset) and external parameters (such as rotation matrix, translation vector). Camera calibration is typically done by capturing images of a known geometric shape calibration board or object, and then calculating the camera's calibration parameters based on the corresponding feature points on the calibration board and in the images. Next, an on-board camera is used to capture images of the road and surrounding environment. Then, extract features from the images using feature extraction algorithms, such as landmarks, lane lines, etc., that can be used for localization [80]. Then, the extracted features are matched with the map or known features to find the corresponding positions of the features in the images with respect

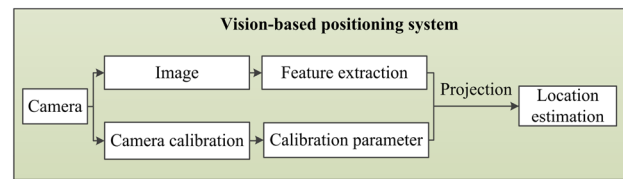


Fig. 6 The framework of vision-based positioning system

to the actual environment. Convert the image coordinates of the matched features to world coordinates. Finally, the position of the vehicle is estimated based on the matched features and the coordinate transformation results, combined with the camera calibration parameters. Among the methods of visual localization, feature-based localization and deep learning-based localization are commonly used. With the continuous development of computer vision, sensor technology, and deep learning, the accuracy and reliability of visual localization will be enhanced.

There are many factors that influence the accuracy of visual positioning, mainly include weather changes, vehicle movement, camera installation location, etc. Weather changes affect the environment of the light intensity changes in the situation, which leads to overexposure or underexposure of the camera phenomenon. The magnitude of vehicle motion speed is inversely proportional to the quality of visual imaging. Due to the limitation of the camera frame rate, when the vehicle speed is large, the captured image will have a certain degree of blurring, losing texture features or generating erroneous textures, which affects the feature shapes and reduces the overlap rate of two neighboring frames. The camera mounting position also affects the visual localization accuracy. The estimation accuracy is reduced due to external calibration of the mounting attitude angle and spatial position offset errors. Generally, fusing the vision information with the information from other sensors to improve localization accuracy.

In [39], a vehicle localization scheme consisting of inertial navigation, binocular camera, and monocular camera was proposed to estimate vehicle location in a tunnel. When there is no GPS signal, the proposed system provides high-precision vehicle lateral localization and accurately acquire the lane where the vehicle is after a lane change or overtaking. The proposed system utilizes a machine vision approach that employs a monocular camera for lane line identification. Subsequently, it fits the identified lane lines to ascertain the vehicle driving status. In [81], authors proposed a novel vision-based positioning approach which only needs one camera for vehicle positioning in underground tunnels.

Digital map

Digital map contains highly detailed street maps with road markings, traffic signs or the environment data (such as visual features, LiDAR maps, etc.) [9]. Based on digital maps and the location information from sensors, vehicle position can be determined using map matching method.

Map matching is a technology that corrects the positioning error of satellite positioning, trajectory projection positioning or other positioning methods through software methods. The map matching method combines existing road digital map information and vehicle sensor data to achieve accurate matching of vehicle positions. In [40], a synthetic reflectivity map considering the reflectivity of tunnel walls and road surface was established using 3D LiDAR in the tunnel, then vehicle position can be obtained through a correlation matching with the reflectivity map. Factors such as map accuracy, initial positional accuracy, and map matching algorithm affect the positioning accuracy and real-time performance of map matching based positioning methods.

Geomagnetic positioning techniques

The magnetic field of earth is not constant, and there are anomalies when magnetic minerals are present in tunnels. By measuring these anomalies and generating a map of the magnetic field, we can use these measured data to successfully locate the target in the tunnel. In [41], the geomagnetic field based positioning method was applied to estimate the target location. In [82], a magnetic vehicle positioning system was proposed for the GNSS-denied scenarios. The core concept of the proposed positioning system is to extract magnetic fingerprints from geomagnetic field anomalies. Through comparing the measured magnetic field with a magnetic map, vehicle location can be obtained without relying on GNSS signals. The matching algorithm proposed in this paper enables the magnetic vehicle positioning system to achieve a positioning accuracy of 5.14 ms. In [83], researchers proposed a magnetic field-based vehicle localization system, which can achieve precise positioning of vehicle using the magnetic sensor of a smartphone with magnetic sensor calibration and attitude estimation functions. The smartphone was fixed on the dashboard. In the proposed localization system, the sensors were calibrated using the location data from magnetic field database and GNSS before entering the tunnel. Based on the obtained magnetic field sequence data, the vehicle locations can be estimated.

V2X-based positioning techniques

V2X-based positioning technology relies heavily on onboard sensors to achieve highly accurate vehicle positioning by interacting this information with other vehicles and

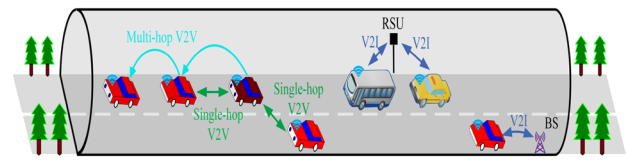


Fig. 7 The V2X-based vehicular positioning. Through single-hop V2V communication and multi-hop V2V communication, the vehicle can obtain localization-related information from surrounding vehicles. Through V2I communication, the vehicle can obtain positioning-related information from the surrounding infrastructures, such as RSU and BS

infrastructure (i.e. BS and RSU) in real time. Currently, the mainstream technical solutions supporting V2X communication include the DSRC technology based on the IEEE 802.11p and 802.11bd protocols, and Cellular Vehicle-to-Everything (C-V2X) technology (including Long Term Evolution (LTE)-V2X and 5G+NR-V2X) [84, 85]. The new study group of IEEE 802.11 Next Generation V2X was formed in March 2018 to develop IEEE 802.11 bd for the DSRC, which is being developed as a WiFi amendment to improve V2X performance [84, 86, 87]. The 3rd Generation Partnership Project (3GPP) has developed C-V2X, the C-V2X standardization work is mainly divided into three phases. Phase 1 is based on LTE technology to meet the basic service requirements of LTE-V2X, corresponding to LTE Rel-14; Phase 2 is based on LTE technology to meet part of 5G+V2X enhanced service requirements (LTE-eV2X), corresponding to LTE Rel-15; Phase 3 is to realize all or most of the 5G V2X enhanced service requirements based on 5G New Radio (5G+NR) technology, corresponding to 5G+NR Rel-16 and Rel-17. Vehicles are equipped DSRC and C-V2X communications equipment, each vehicle can communicate with surrounding vehicles or infrastructures in the communication range to obtain positioning related information via corresponding communication protocols, as shown in Fig. 7.

In [88], authors proposed a vehicle positioning scheme using V2V communication technology. All vehicles are equipped with DSRC transceivers so that each vehicle to be located can communicate with neighboring vehicles within communication range via the IEEE 802.11p protocol. In [89], authors suggested a vehicle-infrastructure cooperative positioning (VICP) scheme considering the RSU perception information and ego vehicle information. In the proposed scheme, the onboard unit (OBU) was used to acquire the RSU perception information via C-V2X communication. A Particle Filter (PF)-based data fusion framework was proposed in [42], which considers both V2V and V2I range-dependent measurements. A pseudolite vehicle positioning system was proposed for tunnel environments in [28]. The proposed positioning system realizes three-dimensional vehicle positioning, which consists mainly of BSs, user receiver, laser ranging module, and digital elevation model (DEM). The BSs transmit ground-based GNSS-like navigation signals

containing timing information. The user receiver receives these navigation signals from the BSs and provides low-latency positioning using both carrier phase and pseudorange measurements. The DEM is used to obtain the priori height information of the user receiver. The laser ranging module is used to measure the distance between the vehicle and the tunnel sidewall. In addition, IMU was used in the proposed positioning system to improve the reliability of vehicle positioning. In the proposed system, C-V2X communication technology is utilized for communication between vehicles and BSs.

Table 2 compares the positioning algorithm, advantages and disadvantages of the commonly used positioning technologies in the literature. Along with the recent evolution of wireless networks termed New Radio, communication-based positioning strategies, especially V2X communication technology, show promising prospects in the field of vehicle positioning in tunnels. V2X communication technology enables real-time communication and exchange of location information between vehicles, which is expected to improve the accuracy and real-time performance of vehicle positioning.

Positioning methods

In the tunnel environment, there are several methods can be used to estimate the location of vehicles. The positioning methods include range-based localization methods, the range-free localization methods, and the multi-sensor fusion localization methods, CP methods, etc. In the following, these positioning methods will be thoroughly described.

The range-based localization methods

The range-based localization methods are utilized to calculate the unknown location of the target vehicle by measuring the distance or angle information between the target vehicle and another nodes using trilateration, triangulation, LS, maximum likelihood estimation, Bayesian inference method, or various filters, etc. These measurements can be obtained by extracting the information contained in the received signal, such as ToA, TDoA, AoA, RSS, etc [90].

ToA

The ToA-based method is based on the calculation of the signal travel time from the transmitter to the receiver when both of them need to have accurately synchronized time clocks. Based on the propagation time and the speed of the radio signals, the distance between the receiver and transmitter can be estimated. The distance can be written as $d = c * t$, where d is the straight distance from the transmitter to receiver, c is the

speed of light; t is the propagation time, namely the measured ToA. Figure 8 shows an example of the ToA-based positioning method using three BSs/access points (APs), which takes the BS/access point (AP) as the center of the circle, and the distance between the BS/AP and measured object as the radius can draw multiple circles, and the intersection of multiple circles is the actual position of the measured object. The following equations can be obtained:

$$\begin{cases} d_1 = \sqrt{(x - x_1)^2 + (y - y_1)^2} \\ d_2 = \sqrt{(x - x_2)^2 + (y - y_2)^2} \\ d_3 = \sqrt{(x - x_3)^2 + (y - y_3)^2} \end{cases} \quad (1)$$

where d_1 denotes the distance from the target node to BS1/AP1, d_2 denotes the distance from the target node to BS2/AP2, d_3 denotes the distance from the target vehicle to BS3/AP3, these distances can be calculated from ToA measurements; the position of the target node is (x, y) , the locations of BS1/AP1, BS2/AP2, BS3/AP3 are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , respectively.

In order to compute the target location in two-dimensional (2D), at least three BSs/APs should be used. This is because a single ToA measurement can only determine the coordinates of a target in a sphere. Similarly, in order to estimate the 3D position of the target, at least four BSs/APs are needed. ToA-based positioning methods requires the clocks at the BSs/APs and the targets to be properly synchronized with respect to absolute time [11]. The mathematical model of this positioning method is based on the assumption of unobstructed propagation of electromagnetic waves, when the electromagnetic wave propagation path contains paths such as reflection or diffraction, the calculated distance will contain errors, such errors are also known as Non Line-of-Sight (NLoS) error. Due to NLoS propagation and other factors, the three circles may not ultimately intersect at a single point, and the corresponding algorithm is needed to estimate the target position, which also introduces some error. In order to improve positioning accuracy, combining all the ToA measurements, the target location can be obtained using least-squares, EKF, PF, maximum likelihood (such as Chan algorithm), etc. Additionally, in practical applications, multiple types of information (such as doppler shift, Direction of Arrival (DoA), etc.) may be integrated. By fusing this information, more accurate and reliable localization results can be achieved. In [91], authors proposed a novel multi-epoch and multi-antenna (MEMA) ToA method to obtain the vehicle position in harsh scenarios. In the proposed positioning method, the authors utilized ToA measurements from multiple recent epochs, introducing inter-epoch constraints on position and attitude changes. The method consists of two steps: first, developing a new MEMA semidefinite programming approach to obtain a reliable initial estimate, and

Table 2 Summary and comparison of vehicle positioning technologies

Technology	Category	Measurements	Algorithm	Advantages	Drawbacks
Radio-based	Pseudo-satellite devices	ToA	ToF ranging algorithm, the trilateration positioning algorithm	High accuracy; unaffected by LoS obstructions	System coverage is limited to the location of the pseudo-satellite transmitter, which is unsuitable for applications requiring extensive localization and is susceptible to signal interference
	UWB	ToA, TDoA, AoA, RSS	Trilateration; triangulation	High accuracy	Limited signal penetration; limited coverage; environmental interference; higher power consumption
	RFID	RSS, Phase-Difference-of-Arrival (PDoA), AoA	Distance fading model; fingerprinting; triangulation	Moderate to high accuracy; low cost; low power consumption	Low positioning accuracy; limited coverage; environmental interference; short range
	WiFi	RSS	Fingerprinting	Medium range (100 to 300 m (WAVE)); low cost	Low accuracy; signal interference; higher power consumption; environmental interference
	Zigbee	RSS, TDoA	Fingerprinting, trilateration	Low power consumption; low cost	Limited data transfer rate; limited communication range; low accuracy; environmental interference; limited network capacity
	Bluetooth	RSS, AoA	Fingerprinting, triangulation	Low cost; low power consumption; small size	Limited positioning accuracy; limited coverage; signal interference; short range
	VLC	RSS, TDoA, ToA, AoA, PDoA	Fingerprinting, proximity, trilateration, multilateration, image transformation	Low cost; immune to electromagnetic interference	Limited coverage; susceptible to light and environmental conditions; LoS required
	LiDAR	ToF	Laser positioning and map matching; filter-based algorithm; machine learning; SLAM	High accuracy; omni-directional sensing; suitable for a wide range of environments; high performance in real time; long range; unaffected by color and texture	High cost; LoS required; limited resolution. Most lidars are fixed in the horizontal direction and can only scan a single plane. Multiple LiDARs or mechanical scans may be required to obtain 3D information; LiDAR requires the target to be reflective, so it may not work properly on non-reflective or irregular surfaces

Table 2 continued

Technology	Category	Measurements	Algorithm	Advantages	Drawbacks
INS-based	mmWave radar	ToF, AoA	Multipath-assisted localization; correlative scan matching; SLAM	High resolution; strong anti-interference ability; long detection range, independent of weather	High cost; fast signal attenuation; large influence by obstruction; high complexity of data processing
	Ultrasonic radar	ToF, TDoA	Trilateration	Strong penetration ability; low cost; no interference from electromagnetic effects	Limited measurement accuracy; small detection angle, short detection distance. Environmental interference, suitable only for low speed; LoS required
	IMU	Acceleration, angular velocity	DR	INSs have less impact on the external environment, and do not need to rely on external BSs or satellite signals	Errors accumulate over time; INS may require an initial alignment to calibrate the sensors and minimize drift errors before beginning use; INS does not provide global position information
Visual-based	Camera	Images	Feature-based positioning, deep-learning localization, SLAM	High accuracy. It can provide real-time location information and rich information about the surrounding environment, such as object recognition, map construction	Visual localization is impacted by many factors, for example, ambient lighting, weather conditions, occlusions, etc. It usually requires significant computational resources
Geomagnetic field-based	Magnetic sensor	Magnetic field strength and direction	Fingerprinting	Good stability and reliability; unaffected by shading; low power consumption; low cost	Limited accuracy; geomagnetic positioning requires the installation of geomagnetic sensor equipment in tunnels; susceptible to geological and topographical influences; geomagnetic sensors require periodic calibration to ensure accuracy
V2X-based	RSU, BS	ToA, TDoA, AoA, PDofA, Doppler shift	EKF, Unscented Kalman Filter (UKF), PF, Unscented Particle Filter (UPF)	High real-time; high reliability	High cost. Appropriate infrastructure, such as RSUs, BSs, etc., needs to be placed. Building these infrastructures may face deployment difficulties in a tunnel environment. High algorithm complexity

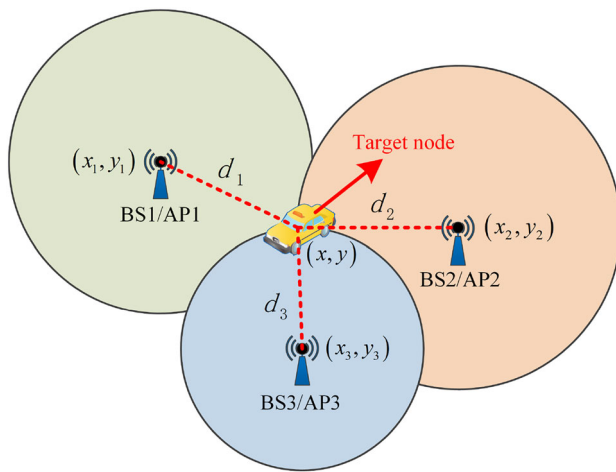


Fig. 8 The ToA-based positioning using three BSs/APs

second, employing an iterative algorithm to refine these estimates based on the initial guess. Simulation results indicate that the proposed MEMA ToA method achieves high localization accuracy and robustness. In [92], a novel vehicle positioning approach utilizing ToA and doppler shift measurements was proposed. Then EKF and KF were used to estimate vehicle position. Results show that the EKF performs better than KF and the average estimation error is reduced by 10m.

TDoA

The TDoA method locates the target node by measuring the arrival time difference of signals between the BSs/APs and the target node. Signals sent from the target node reach at least three synchronized receiver antennas situated at different locations with varying arrival times. The location of target node can be determined by computing the time difference of the received signals at the respective antennas [11].

When the TDoA-based localization method is used to achieve 2D localization of a target, at least three BSs/APs are required. At least four BSs/APs are required to achieve 3D localization of the target node. Figure 9 shows an example of the TDoA-based positioning method using three BSs/APs. The basic principle of TDoA-based localization: firstly, measure the time difference t_{21} from the target node to the BS1/AP1 and BS2/AP2, which is converted into the distance difference R_{21} , $R_{21} = c * t_{21}$, where c is the speed of light; then the target node must be located in the hyperbola with BS1/AP1 and BS2/AP2 as the focus, and the difference of the distance to the focus is always R_{21} . Then measure the distance difference R_{31} from the target node to BS1/AP1 and BS3/AP3, then the target node must also be located in the hyperbola with BS1/AP1 and BS3/AP3 as the focus, and the distance difference to the focus is always R_{31} , $R_{31} = c * t_{31}$,

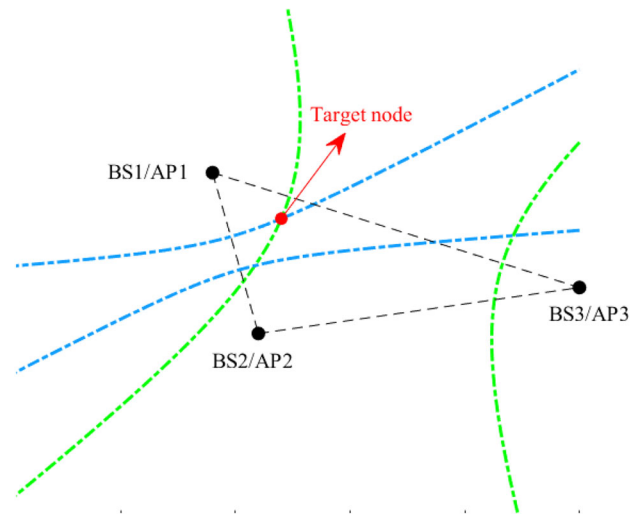


Fig. 9 The TDoA-based positioning using three BSs/APs

where t_{31} is the time difference from the target node to the BS1/AP1 and BS3/AP3. The following equations can be obtained:

$$\begin{cases} \left(\sqrt{(x-x_2)^2 + (y-y_2)^2} - \sqrt{(x-x_1)^2 + (y-y_1)^2} \right)^2 = R_{21}^2 \\ \left(\sqrt{(x-x_3)^2 + (y-y_3)^2} - \sqrt{(x-x_1)^2 + (y-y_1)^2} \right)^2 = R_{31}^2 \end{cases} \quad (2)$$

where the locations of all the anchors are known, the locations of BS1/AP1, BS2/AP2, BS3/AP3 are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , respectively. The location of the target node to be estimated is (x, y) . The coordinates of the intersection of the two hyperbolas are obtained by solving the above system of equations in Eq. (2), there will be two points of intersection of the two hyperbolas. Based on the a priori information one can be excluded from the computation process, and the other intersection is the coordinate of the target node. The target node is located at the intersection of the three hyperbolas. In real-world environments, signal transmission paths are subject to various interferences such as multipath effects, signal attenuation, and environmental noise, leading to errors in the measured time differences. Due to these measurement errors and noise, the set of equations obtained from the TDoA-based localization method may not have a unique solution or may lack sufficient accuracy. In such cases, the position of the target cannot be estimated by directly solving the intersections. Instead, other localization algorithms (such as least squares, Kalman filtering, particle filtering, etc.) are typically used to reduce the impact of measurement errors, suboptimal base station geometry, and noise, thereby improving localization accuracy.

Compared to ToA-based localization method, TDoA-based localization method does not require high-precision time synchronization between BSs/APs and the targets. The TDoA-based localization method requires high-precision

time synchronization between BSs/APs. Multiple receivers (also known as BSs or receivers) need to operate on the same time reference to accurately measure the time difference of signal arrival. This time synchronization can be achieved through high-precision clocks, such as GPS timing or other advanced synchronization technologies. In [93], authors proposed a positioning method to calculate the vehicle location in the tunnel using ToA and TDoA information. Authors verified the effectiveness of the proposed positioning method through simulation using SUMO (traffic simulator) and NS2 (network simulator). In [56], based on traditional UWB, a novel vehicle positioning method was proposed to obtain accurate vehicle locations in the tunnel, utilizing the down-link TDoA information.

AoA

The angle-based localization techniques utilize either the AoA, the Angle of Departure (AoD), or even both. The AoA measurement determines the radio frequencies propagation direction or high signal power in a series of an antenna. The target vehicle's location is determined by computing the intersection of two angular lines. Each of these lines is formed by the radius originating from one main station [11]. In order to obtain a high localization accuracy, at least two anchor nodes are required for 2D localization. At least three anchor nodes are required for three-dimensional (3D) localization. The AoA localization approach does not require clock synchronization, it requires complex hardware to determine the AoA. Figure 10 illustrates a schematic of the AoA-based positioning using two BSs/APs. The following equations can be obtained:

$$\begin{cases} \tan \alpha_1 = \frac{y - y_1}{x - x_1} \\ \tan \alpha_2 = \frac{y - y_2}{x - x_2} \end{cases} \quad (3)$$

where the location of the target node to be estimated is (x, y) , the locations of BS1/AP1 and BS2/AP2 are (x_1, y_1) and (x_2, y_2) , respectively. The AoA of the target node to BS1/AP1 and BS2/AP2 are α_1 and α_2 , respectively. From Eq. (3), we can obtain:

$$\begin{bmatrix} y_1 - x_1 \tan \alpha_1 \\ y_2 - x_2 \tan \alpha_2 \end{bmatrix} = \begin{bmatrix} -\tan \alpha_1 & 1 \\ -\tan \alpha_2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad (4)$$

We can write Eq. (4) in matrix form, it can be written as

$$\mathbf{Y} = \mathbf{A}\mathbf{X}, \quad (5)$$

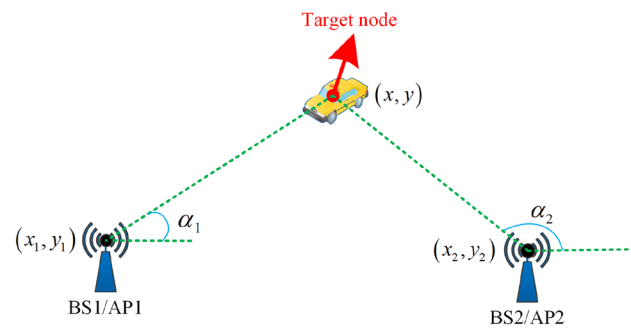


Fig. 10 The AoA-based positioning using two BSs/APs

where

$$\mathbf{Y} = \begin{bmatrix} y_1 - x_1 \tan \alpha_1 \\ y_2 - x_2 \tan \alpha_2 \end{bmatrix}, \quad (6)$$

$$\mathbf{A} = \begin{bmatrix} -\tan \alpha_1 & 1 \\ -\tan \alpha_2 & 1 \end{bmatrix}, \quad (7)$$

$$\mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}. \quad (8)$$

Based on the AoA measurements, the location of the target node can be obtained by solving Eq. (5) using LS algorithm. The effects of measurement error and noise are not considered in Eqs. (3), (4), and (5). However, AoA measurements are often affected by various sources of error, including multipath effects and sensor inaccuracies, which significantly increase the uncertainty of AoA measurements. In AoA-based positioning methods, due to the influence of measurement errors and noise, the system of equations for positioning may lack a unique solution or have insufficient accuracy. To effectively utilize AoA measurements for target localization, nonlinear optimization techniques (such as the Levenberg-Marquardt algorithm), PF, EKF and UKF are typically employed to better handle nonlinear characteristics and measurement errors, thereby enhancing positioning accuracy and reliability.

In [94], authors proposed a novel DoA and vehicle positioning strategy based on the exact spatial geometry. In the proposed positioning strategy, firstly, a two-stage unitary approximate message passing (UAMP)-based method was developed to estimate DoA; then based on the estimated DoAs at multiple BSs or RSUs, the location of each vehicle can be obtained with cross-localization criterion. In [95], a novel vehicle positioning strategy using DoA information was proposed, where a dual one-dimensional (1D) Convolutional Neural Network (CNN) under DL framework was developed to obtain the 2D DoA estimation of incoherently distributed (ID) sources, and the received signal to noise ratio (SNR) data was used to select the best set of DoA estimates from multiple BSs for achieving high-precision vehicle positioning. In [96], a positioning strategy using DoA

and ToA was proposed. A low-complexity Estimating Signal Parameters via Rotational Invariance Techniques (ESPRIT) algorithm based on total least squares was proposed to estimate DoA and ToA information, and the UKF was used to calculate the vehicle location based on the estimated DoA and ToA parameters.

RSS

The RSS-based localization techniques use the values of received signal strength and the radio signal propagation loss model for distance measurement and localization estimation. The basic steps of this localization method are as follows: first, measure the signal strength between the target mobile device (such as vehicle) and the base station. Signal strength can be represented by parameters such as Received Signal Strength Indicator (RSSI) and RSS. Then, based on the signal propagation model (such as free space propagation model or logarithmic distance attenuation model), combined with the measured signal strength, the distance between the target mobile device and the base station/AP can be estimated. The distance between the target mobile device (i.e., vehicle) and BSs/APs can be written as [97]:

$$d = d_0 * 10^{\left(\frac{RSS_0 - RSS}{10 * n}\right)}, \quad (9)$$

where d is the distance between the target mobile device and BS/AP, d_0 is the reference distance, the RSS value at the reference distance d_0 is RSS_0 , RSS denotes the measured signal power for the received BSs or APs signals, n is the path loss exponent for the received BSs or APs signals. Finally, based on the distance information between multiple BSs/APs and the target mobile device, and the known positions of the BSs/APs, algorithms such as trilateration positioning can be used to calculate the position of the target mobile device relative to the fixed BSs. To estimate the 2D coordinates of a target mobile device, a minimum of three BSs or APs is required. For estimating the 3D coordinates of the target mobile device, at least four BSs or APs are necessary.

In [98], authors used a novel RSSI based localization method to compute the vehicle location in tunnel environment, and the long term evolution signals were obtained using mobile phones. In [99], a localization algorithm leveraging the linear modified log function model for RSSI was introduced. Experimental results demonstrated enhanced accuracy and anti-interference capabilities in RSSI localization through the utilization of the proposed algorithm based on linear modified log function. Many factors can affect the power of the received signal, such as shadow fading, multipath fading, path loss, and blocking, etc. These factors impact the accuracy of RSS-based positioning. To address the errors in RSS-based positioning, some researchers have proposed

the fingerprinting method. The details of the fingerprinting method will be elaborated in Sect. 4.2.

Hybrid localization methods

To enhance the accuracy of vehicle localization, multiple measurements are fused to calculate the location of the target vehicle. In [92], authors proposed a novel solution for vehicle localization using ToA and Doppler shift (DS) measurement datas. The localization method presented in this paper combines the velocity and initial GNSS location of the vehicle as it enters a tunnel, then uses Doppler shift and ToA information to track the vehicle inside the tunnel. In [100], a single-site vehicle localization method was proposed to locate vehicle in the rectangular tunnel environment. The proposed positioning method using the ToA and DoA data of reflection paths from two tunnel walls to locate the target vehicle. However, the author validated the effectiveness of the proposed localization algorithm solely through theoretical analysis and simulation experiments, without applying it to real tunnels.

The range-free localization methods

The range-free localization method can be implemented using a variety of techniques and sensors. These methods rely on not directly measuring the distance between the vehicle and a landmark or reference point, but rather using other information to estimate the vehicle positions. The commonly used range-free positioning method is fingerprinting localization method. Fingerprinting localization is a technique that relies on the creation of a fingerprint or unique signature of the environment. The main fingerprint types include motion fingerprint, visual fingerprint, and signal fingerprint, where signal fingerprint is widely used [101]. The signal fingerprint-based localization methods often use wireless signals, such as WiFi, Bluetooth, RFID, UWB, Zigbee, cellular signals, or geomagnetic field signal, etc.

Figure 11 illustrates the framework of fingerprint-based vehicle positioning method. The fingerprinting localization method includes two phases: offline phase and online phase [97]. During the offline phase, fingerprint data is collected and a fingerprint database is created. First of all, a database of signal strength measurements is created by surveying the environment. This involves taking measurements of received signal strengths or magnetic fields from various APs or base stations at different known locations within the area of interest. Then, the collected measurements are then stored in a database, associating each set of measurements with its corresponding known location. During the online phase, if the vehicle needs to determine its location within the environment, it measures the signal strengths of nearby access points or BSs. Then, many localization algorithms can be

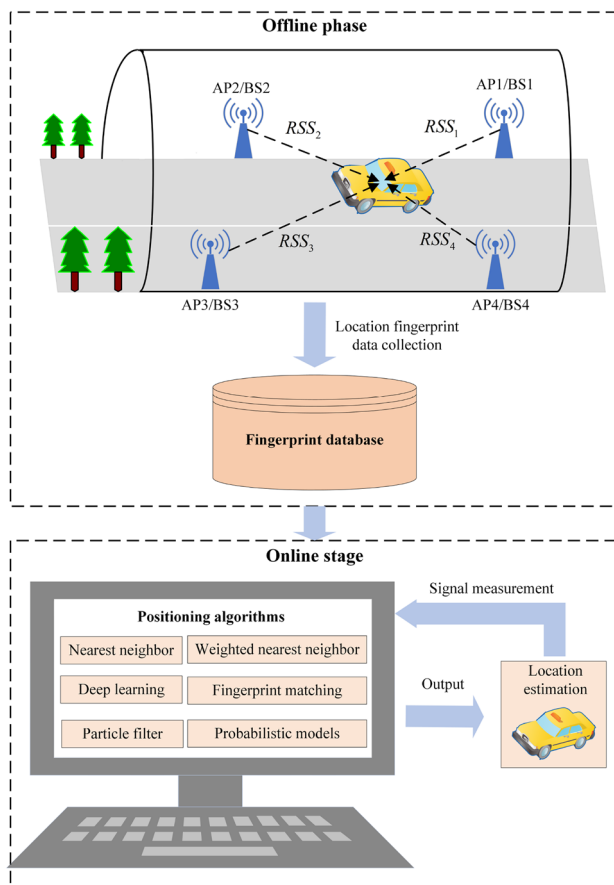


Fig. 11 The fingerprint-based positioning method

used to achieve vehicle positioning, such as nearest neighbor, weighted nearest neighbor, nearest neighbor interpolation, deep learning, probabilistic models, fingerprint matching, PF, EKF, etc. For the fingerprint matching method, the measured signal strengths are compared to the database of fingerprints, and a best match is found. The known location associated with the best match in the database is then considered to be the current location of the vehicle.

In [102], authors introduced an indoor vehicle positioning method that combines deep learning and frequency modulation (FM) fingerprint mapping. This approach involves reconstructing the FM fingerprint map through a specialized deep CNN. Subsequently, the vehicle location can be acquired using a continuous vehicle positioning algorithm that takes into account the vehicle movement status. This positioning strategy also applies to vehicle positioning in tunnels. In [103], authors used the Parabolic Wave Equation (PWE)-based wireless channel simulator to acquire the DoA information for the construction of fingerprint dataset. The Multiple Signal Classification (MUSIC) algorithm and PWE methods were employed to estimate DoA information. Then fingerprint localization algorithms (i.e., Weighted K-nearest Neighbour (WKNN) and K-nearest Neighbour (KNN) algo-

rithms) were utilised to estimate the vehicle positions. In [35], authors used WiFi fingerprint positioning method to estimate the target position. To enhance the precision of WiFi-based positioning within tunnels, authors suggested diverse access point layout strategies tailored to various tunnel environments. They created a dynamic fingerprint database utilizing RSSI vectors from designated test points. Additionally, the authors introduced a positioning algorithm, KNN-FCM, which combines the KNN algorithm with Fuzzy C-means Clustering (FCM) to calculate the target position.

In [104], a localization method for ZigBee-based location fingerprints, utilizing an enhanced K-means algorithm, was introduced to determine the whereabouts of personnel within the tunnel. The proposed localization method comprises two stages: offline and online. During the offline stage, RSSI data is gathered, while in the online stage, the mobile node compares real-time RSSI data received from each fixed node with the stored location fingerprint data in the database. This process was executed sequentially to identify the reference node with the closest RSSI Euclidean distance. Subsequently, the localization algorithm was applied to calculate the location of target node and output the final localization result.

Fingerprinting localization can be quite accurate, especially in the environments where GNSS signals are weak or unavailable. However, this positioning method has some limitations. For example, the changes in the environment (such as moving objects or structural modifications) can affect the accuracy of fingerprinting over time. Signal strength can vary due to interference, obstacles, or the presence of people, leading to inaccuracies. There are often other vehicles or obstacles in tunnels that may interfere with the propagation of the wireless signal, thus affecting the accuracy of the positioning. Besides, signal reflection, refraction, and multipath effects are often present in tunnels, and these phenomena may lead to instability in wireless signal strength and reduced positioning accuracy. The database of fingerprints needs to be regularly updated to account for changes in the environment. Tunnels with complex environments, as well as long tunnels, may result in large database sizes that become more complex to manage and maintain, as well as potentially introducing more errors. Fingerprint-based localization methods rely on pre-existing wireless infrastructure, such as Wi-Fi routers or Bluetooth beacons. If these infrastructures are insufficient or unavailable in the tunnel, the effectiveness of the localization method will be limited.

The multi-sensor fusion based localization methods

Many sensors have their limitations. Usually, when vehicle localization is performed based on individual information source, the accuracy sometimes cannot meet the requirements. In order to enhance vehicle localization, navigation, and sensing capabilities, the method of fusing information

from multiple sensors or data sources is usually required. Multi-sensor fusion positioning refers to the use of data from multiple sensors installed on a vehicle, such as an IMU, GPS, camera, and Lidar, to improve the accuracy and robustness of vehicle positioning. This approach focuses on fusing data from different sensors to enhance positioning performance through fusion processing. The common methods for multi-sensor data fusion include filter-based method (such as UKF, EKF, UPF, PF, adaptive filter, Kalman Filter (KF), etc.), FG based method, machine learning-based, etc.

The filter-based method

The linear KF is capable of achieving an optimal estimation of the target's state within the constraints of a linear Gaussian model, resulting in improved tracking performance. However, the internal system exhibits varying degrees of non-linearity. In nonlinear systems, the state transfer function and observation function are generally nonlinear and the standard KF algorithm cannot be directly applied.

EKF is an improvement on KF and is suitable for dealing with nonlinear systems. The fundamental concept of the EKF is to approximate a nonlinear system through linearization, allowing the application of KF for state estimation. EKF is a commonly used sensor fusion algorithm that fuses measurements from different sensors into a consistent state estimate for target localization and navigation. In the tunnel, EKF can combine data from multiple sensors (i.e., GPS, inertial sensors, radar, and camera, etc.) to improve vehicle positioning accuracy. In [105], authors proposed a modified EKF algorithm that utilizes past available measurements and sensor measurements (such as IMU measurements) to estimate vehicle position in GNSS-denied environments. The authors carried out measurements in two scenarios, a sea tunnel and a extreme dense environments of Hong Kong city, to evaluate the performance of the proposed algorithm. In [38, 69, 72], the EKF was employed to fuse multi-sensor information. EKF employs a linearization approach, which transforms the original problem into an approximated problem and solves it optimally, rather than approximating the solution of the correct problem. This approach may significantly impact solution accuracy or potentially result in system divergence. Hence, when dealing with highly nonlinear and non-Gaussian noisy systems, more sophisticated nonlinear filtering techniques can be utilized, such as UKF, PF, and exact-recursive nonlinear filters. These filters preserve the nonlinear structure problem and can effectively enhance the system performance [106].

The UKF is the combination of KF and unscented transformation (UT), which is more accurate than Taylor series expansion, and can effectively overcome the problems of poor stability and low estimation accuracy of EKF. UKF is highly accurate in calculating nonlinear distribution statistics

because it does not need to ignore the higher-order terms. The UKF has the following disadvantages, such as for linear systems, the filtering effect of UKF is the same as KF; for nonlinear systems, the filtering performance of UKF is better than that of EKF; and there is no need to compute the Jacobi matrix (it is also suitable for the occasions where it is difficult to obtain the Jacobi matrix). Of course, UKF also has some drawbacks, such as the computation speed is slower than EKF, the filtering effect is better for Gaussian distributions. In addition, the filtering effect is worse for non-Gaussian distributions with high confidence. Few scholars have used this filter for fusion of multi-sensor information in tunnel scenarios.

The PF algorithm has extensive application in addressing parameter estimation and state filtering challenges within nonlinear systems and time-varying systems with non-Gaussian noise. The PF algorithm has the problems of degradation of particles and poor real-time performance. In [42, 107, 108], the PF was utilized to fuse the information from various sensors, including inertial sensors and digital map, radar, etc. Results indicated that the positioning method based on filtering exhibits superior performance.

The UPF algorithm combines the advantages of UKF and PF. The UPF avoids the problem of particle degradation as much as possible by propagating the mean and covariance information of the state to the particle ensemble by using the UT. UPF is able to better capture the nonlinear properties of the system state and improves the filtering performance of PF. In addition, the UPF algorithm avoids linearization of nonlinear functions, it is usually more robust than UKF in dealing with highly nonlinear systems. In [109], the UPF was used to fuse the data from different localization sources, specifically, GPS, BeiDou and IMU. In [100], authors employed the UPF to process the obtained ToA and DoA data, and achieved the vehicle localization in the rectangular tunnel scenario.

The adaptive filter automatically adjusts the weights according to the reliability and accuracy of the sensor data to enhance the location precision. This is useful for dealing with unstable sensor data in tunnels. In [78], the adaptive Kalman filtering method known as Motion Inertia Estimated Adaptive Kalman Filtering (MIAKF) leverages the linear physical characteristics of tunnels. It dynamically adjusts fusion weights for UWB and IMU based on historical motion states. This adaptive fusion of UWB and IMU information not only alleviates the signal impact from the tunnel environment but also addresses challenges related to IMU initial alignment and error accumulation.

Factor graph-based method

The factor graph is an innovative method for fusing information from multiple sources, with versatile applications in the domains of SLAM, navigation, and positioning. In

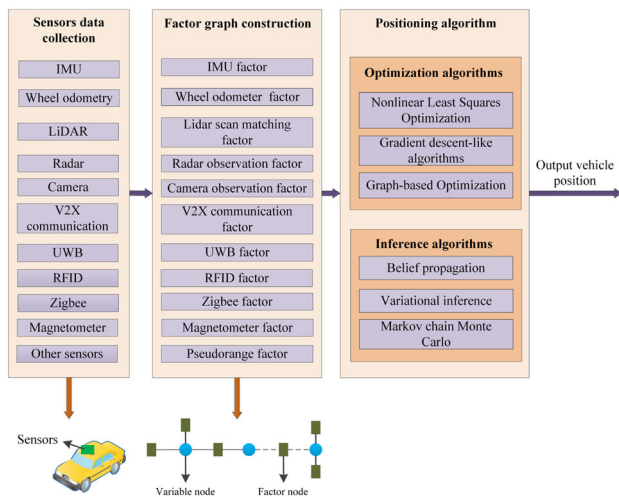


Fig. 12 A factor graph-based framework for vehicle localization in tunnels

recent years, multi-sensor fusion vehicle localization methods based on FGs have been applied to some very challenging scenarios, such as high buildings, dense traffic scenarios, tunnels, etc [110–114]. A factor graph-based framework for vehicle localization in tunnels is shown in Fig. 12. The vehicle localization method based on FG relies on the principles of probabilistic graphical models. It constructs a graph structure to represent the vehicle localization problem and employs an optimization process to estimate vehicle position. This localization method typically involves an optimization problem with the objective of minimizing errors throughout the entire factor graph, ensuring that the estimated vehicle position aligns with all sensor information and constraints. The Factor Graph Optimization (FGO) process can be implemented using algorithms such as gradient descent. By modeling sensor information, constraints, and error terms, the FG-based vehicle localization method efficiently handles multiple sensor inputs, enhances localization accuracy, and exhibits a degree of real-time capability and adaptability.

Currently, some authors used FG for vehicle localization in tunnel scenarios [115, 116]. In [115, 116], the sliding window based factor graph method was used to fuse the data obtained from IMU, camera, LiDAR, rail vehicle wheel odometer and GNSS. In [38], the authors used FG-based positioning method to mitigate the NLoS error of radio signal. In [117], a FGO-based Multi-GNSS Real-time Kinematic (FGO-RTK) strategy was developed to achieve precise positioning in urban canyons. Moreover, the effectiveness of the FGO-RTK localization algorithm was verified in medium urban and deep urban scenarios.

Machine learning-based method

With the development of machine learning, significant progress has been made in vehicle tracking methods based on various machine learning algorithms [23]. Many studies are dedicated to leveraging machine learning techniques to fuse measured data acquired from various sensors, for example, GNSS receivers, IMU, LiDAR, RFID, and visual sensors, thereby enhancing the robustness and accuracy of vehicle localization.

The basic principle of machine learning-based vehicle positioning algorithms is as follows: First, we need to collect data generated by various sensors installed on the vehicle, such as IMU, cameras, etc. These data include information like position, velocity, acceleration, and direction, and may also encompass surrounding environmental sensor data, such as camera images. Second, we can extract features from the collected raw data to convert it into a format suitable for machine learning models. The goal of feature extraction is to capture useful characteristics from the data that describe the vehicle's state and its surroundings, such as time-series features, spatial features, and spectral features. Third, we need to preprocess the extracted features, including tasks like data cleaning, normalization, and noise reduction, to ensure the quality and stability of the data input to the model. Fourth, we can select an appropriate machine learning model, such as neural networks, decision trees, or support vector machines, and train it based on the data characteristics and the requirements of the positioning task. During training, the model learns from a large amount of labeled data (data with known real-world position information) to infer the accurate position of the vehicle from the features. Fifth, the trained model is optimized and hyperparameters are adjusted to improve the generalization ability and localization accuracy of the model. Finally, when the model training is completed and validated, it is deployed into an actual vehicle positioning system. The system inputs the trained model based on real-time collected sensor data, such as IMU data, and camera images, to predict and update the vehicle position information. Additionally, we need to regularly update and optimize the machine learning model based on feedback and new data from real-world applications to adapt to changing environments and improve positioning accuracy.

In [118], authors proposed a vehicle positioning system based on V2I communication using CNN. A innovative approach based on deep learning for vehicle localization in indoor environments was introduced in [119]. The proposed method can directly learn the vehicle motion state from the raw data gathered by multiple sensors, using the smartphone built-in sensor data. Subsequently, the trained lightweight neural network was applied to forecast the velocity and heading of the vehicle. In the paper, the long short-term memory (LSTM) networks based positioning method and

the MobileNet-V3 based positioning method were used to achieve efficient vehicle positioning for the situations demanding high positioning accuracy and the situations demanding limited computational resources, respectively. In [52], authors proposed a novel Least Square Support Vector Machine (LSSVM) algorithm to calculate the distances between the RFID tags and the RFID reader based on RSS, and developed a LSSVM multiple model filter algorithm to integrate data from in-vehicle sensors and RFID and estimate the vehicle position in tunnels.

Cooperative localization methods

In addition to the multi-sensor integration, CP approach is also widely used to address the shortcomings of GNSS and obtain vehicle locations in many GNSS-denied environments [120]. CP refers to the process in which multiple positioning devices (e.g., sensors, BS, satellite navigation systems, communication devices, etc.) improve positioning accuracy and reliability through mutual cooperation and information sharing [121].

The basic principle of the cooperative vehicle localization method is to share their respective position and sensor information through communication interactions between multiple vehicles or with the infrastructure, and then use this information for mutual correction and fusion to improve the accuracy and reliability of localization. This approach relies on collaboration between vehicles and solves the problem that a single node in a traditional positioning system is susceptible to occlusion and multipath effects by means of cooperative positioning algorithms and data fusion techniques. Along with the VANET and wireless communication technologies development, V2X-based cooperative positioning has become an important development trend for vehicle positioning in tunnels. In [89], a V2I-based cooperative positioning method was proposed for GNSS-denied scenario, which considers the roadside perception and ego vehicle information. A cooperative neighboring vehicle positioning system was presented in [122], estimating the vehicle locations based on the information obtained through various sensors. In [120], a robust cooperative positioning approach based on the tight integration between GPS pseudo-range and UWB ranging measurements using robust KF was presented for the visible GPS satellite limited environments. The UWB unit measures the internode ranging between the subject vehicle and neighbouring vehicles or RSUs by V2X communication. In [121], authors proposed a novel cooperative inertial navigation (CIN) method to promote the localization accuracy of INS-based positioning system in difficult GNSS scenarios, such as in tunnels and dense urban areas. For the proposed CIN method, vehicles share their IMU and INS-based location information with the other vehicles, and the information obtained from the neighboring vehicles

are fused with the local observations and the carrier frequency offset of the received packets to improve positioning accuracy. In [88], authors proposed a VANET distributed cooperative positioning scheme for use in tunnels and urban canyons. The proposed positioning scheme uses the Round Trip Time (RTT) technique for inter-vehicle distance estimation and combines it with INS measurements to update the position of target vehicle and neighboring vehicles. In [42], authors proposed a CP scheme mixing V2I and V2V range-dependent measurements. Moreover, in order to promote the positioning accuracy of cooperative localization, authors proposed practical trade-offs between RSU deployment costs (i.e., geometric configuration, density) and V2I technology (i.e., Impulse Radio-UWB or ITS-G5).

Multi-path cooperative vehicle positioning technology refers to the use of multiple sensors and multiple signal paths (such as direct and reflected paths) on a vehicle to improve positioning accuracy and reliability. This technology focuses on utilizing multiple signal paths to enhance positioning performance. Traditional methods for addressing multipath issues involve estimating the channel impulse response (CIR) at a receiver and attempting to mitigate the impact of multipath components (MPCs) on the LoS path. In contrast, multipath-assisted positioning does not aim to counter MPCs but rather to utilize them for positioning purposes. Each MPC reaching a receiver via a distinct propagation path can be treated as a distinct signal sent synchronously from another signal source in a pure LoS condition to the receiver, where the signal source is named as virtual transmitter (VT). Gentner et al. [123] describes a Channel-SLAM algorithm for multipath assisted positioning. The fundamental idea behind Channel-SLAM is to reinterpret multipath signals as virtual direct LoS links originating from additional VTs. These transmitters are considered to be stationary in pieces and their positions are influenced by the geometry and location of nearby reflectors. The Channel-SLAM concept was utilized by Markus Ulmschneider to accomplish vehicle positioning in [124]. However, this approach does not fully utilize the inherent multi-user characteristics of vehicular traffic, treating the localization of each vehicle as completely independent. In [125, 126], authors exploited multi-vehicle nature of typical road traffic and the ability for different vehicles to cooperate with each other, and proposed a multi-vehicle collaborative mapping and positioning approach, which is referred to as Team Channel-SLAM. Results show that the proposed positioning method can enhance the accuracy of vehicle localization in the scenario with multiple vehicles. All these multi-path cooperative vehicle positioning methods can be used to locate vehicles in tunnel scenarios.

Table 3 presents common positioning methods, summarizing positioning algorithms and their reported positioning accuracy in the literature. Additionally, we determine whether measurement experiments were conducted to vali-

date these methods. According to the Table 3, it is evident that the predominant method for vehicle localization in tunnel scenarios is multi-sensor fusion positioning. In tunnel environments, the reliability and stability of V2X communication face challenges, leading to relatively fewer applications of V2X-based cooperative vehicle positioning in these scenarios. However, with the rapid development of vehicular networking and communication technologies, V2X-based CP remains a promising method for vehicle localization.

The challenges of vehicle localization in the tunnel environment

Vehicle positioning techniques in tunnels face a variety of challenges related to signal propagation, environmental conditions, and positioning algorithms. Here, we present an analysis of the common challenges.

Signal blocking and attenuation

Objects, structures and walls in the tunnel may cause attenuation and blockage of the wireless signals, thus affecting the signal strength and making distance measurement more difficult. Due to the shielding of the tunnel walls, satellite signals cannot travel through the tunnel walls, GNSS signals are often very weak or unavailable inside tunnels. This means that traditional GNSS positioning methods cannot be used inside tunnels, and other sensors or technologies need to be employed to compensate for this shortcoming.

Signal blocking and attenuation in tunnels can lead to a reduction in location precision, there are many ways to enhance positioning accuracy in such environments.

- The arrangement of base stations in tunnels can be optimized by increasing the number of BSs, adjusting the direction of antennas, or adding signal relay equipment to improve the signal coverage.
- The multi-sensor fusion technology can be used to compensate for the uncertainty caused by signal occlusion and attenuation to improve the accuracy of positioning [134]. For example, in the case of severe signal attenuation, the INS can be applied to measure the angular velocity and acceleration of the vehicles to estimate the trajectory of the vehicles. The INS can provide accurate position information in a short period of time [4, 121].
- The estimation of relative position can be improved by utilizing V2V communication. Through V2V communication, vehicles can exchange position information and can know the position and motion status between vehicles more accurately.
- GPS signal enhancement techniques can be introduced in tunnels, such as underground enhancement system or dif-

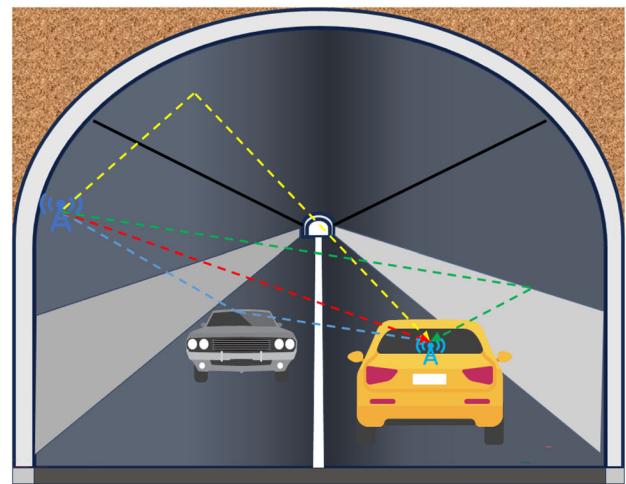


Fig. 13 The multipath propagation in the tunnel

ferential GPS. These techniques can provide more stable GPS signals.

Multipath effect

Multiple paths exist for electromagnetic wave transmission in a tunnel, as shown in Fig. 13. Wireless signals in tunnels may have a multipath effect due to multiple reflection paths, which makes it difficult for the positioning system to accurately discriminate between primary and reflected signals. The presence of multipath causes changes in the signal in terms of delay, amplitude and phase, which results in energy attenuation and a decrease in the signal-to-noise ratio, leading to the fact that the first reached signal is not a direct signal, causing ranging errors and a decrease in positioning accuracy. The effects of multipath will lead to an increase in the complexity of the localization algorithm. Many methods and strategies can be adopted to mitigate the impact of multipath propagation and enhance positioning accuracy [135].

- Utilizing multipath to improve vehicle positioning accuracy. We can adopt some high-precision channel parameter estimation algorithms (such as space-alternating generalized expectation-maximization (SAGE), ESPRIT, Multiple Signal Classification (MUSIC), Kalman enhanced super resolution tracking (KEST), DL, et al.) to extract the multipath signals in tunnel scenarios, effectively utilize these multipath signals, and adopt the multipath cooperative approach for vehicle localization. For example, Markus Ulmschneider et al. proposed a multipath assisted vehicle positioning method for urban environment [124]. In the proposed strategy, the KEST algorithm was used to estimate the parameters of multipath components (MPCs); then, each multipath component arriving at the receiver through different propagation paths can be

Table 3 Comparison of positioning techniques in the tunnel environments

References	Technologies	Data type	Algorithm	Accuracy	Mea.
[5]	Pseudo satellite	ToF	ToF ranging algorithm, trilateral positioning	–	Yes
[28]	Pseudolite devices, IMU, laser ranging module	Velocity, acceleration, pseudorange measurement, carrier phase, the range between vehicle and tunnel sidewall	WLS, digital elevation model	< 11.55 cm	Yes
[53]	UWB	ToA	The exceptional value filtering, the greedy-based clustering algorithm, LS	20.48 cm (average error)	Yes
[54]	UWB	ToA	KF	15–18 cm	No
[55]	UWB	ToA, TDoA	Bilateral two-way ranging method, the differential mapping method of dual UWB locators	8.2% (maximum deviation), 2%	No
[56]	UWB	TDoA	Downlink TDoA technology	Centimeter level	Yes
[78]	UWB, IMU	ToA, velocity, acceleration	MIAKF	The localization errors decrease 10% in mine tunnels.	Yes
[127]	UWB, IMU	Round-trip ToA	An automatic approach for simultaneous refinement of sensors' localization and target tracking	< 5.5 m (RMSE)	No
[38]	UWB, MEMS INS	TDoA, velocity, acceleration	FG-based UWB positioning method, residual weighting algorithm, EKF-based loosely coupled UWB/INS fusion method	0.5397 m (RMSE)	Yes
[46]	RFID, wheel speed sensors, MEMS-based inertial sensors, electronic compasses	RSS, velocity, acceleration, orientation, yaw rate	Least mean square federated filter, the global fusion algorithm based on IMM-STCKF	The straight line driving test: 1.73 m (RMSE), 2.87 m (maximum error). The complex testing environment comprising straight line and curvilinear scenarios: 2.07 m (maximum error), 1.44 m (RMSE)	Yes
[48]	RFID, inertial sensors	RSSI, speed, orientation, acceleration	A matrix-based tag deployment strategy, a collaborative location correction scheme, dead reckoning	When inter-tag interval is 100 m, the RMSE is less than 0.5 m. When inter-tag interval is 1000 m, the RMSE is less than 6 m	Yes

Table 3 continued

References	Technologies	Data type	Algorithm	Accuracy	Mea.
[52]	RFID, in-vehicle sensors	RSS, velocity, acceleration, orientation, yaw rate	LSSVM algorithm, LSSVM multiple model algorithm	The straight-line driving test: 3.17 m (maximum error) and 2.03 m (RMSE). The comprehensive test including straight-line and curvilinear situations: 3.13 m (maximum) and 2.71 m (RMSE)	Yes
[128]	RFID	RSSI	The quadratic weighted centroid algorithm	0.7913 m (average error), 0.6968 m (mean square error), 0.2728 m (standard deviation) and 1.3174 m (maximum error)	No
[129]	RFID	RSSI	PF-based algorithm	In long tunnels without straight feature walls, the positioning error may range from as small as 30cm to as large as several meters	Yes
[35]	WiFi	RSSI	Fuzzy C-means clustering and KNN algorithm (FCM-KNN)	Using a dynamic fingerprint database: 0.94 m (positioning error in rugged and narrow tunnel), 1.01 m (positioning error in long straight tunnel), and 0.80 m (positioning error in wide and narrow tunnel)	Yes
[104]	Zigbee	RSSI	Fingerprint positioning method, improved K-means algorithm	1.05 m (average error)	Yes
[36]	BLE	RSSI	The proximity approach	–	Yes
[37]	LED	Optical signal	Cell-ID positioning method	0.27 m	Yes
[63, 130]	LED	State of optical channel	Trilateration method	< 16.4 cm (average error)	No
[65]	LED, inertial navigation sensors	Optical signal, speed, acceleration	Filter	10 cm (mean error)	Yes
[66]	LED lights, RGB sensors	Color temperature, frequency ratio	Data interpolation	0–2 cm	Yes
[39]	Cameras, inertial navigation sensors, odometer	Heading angle information of vehicle, image	Machine vision method, a lane-level location method based on multi-sensor fusion	–	Yes
[131]	Wheel speed sensor, gearbox speed sensor, accelerometer, gyroscope, GPS receiver, digital map	Speed, acceleration, yaw rate	KF, autoregressive moving average (ARMA) model	< 100 m (positioning error)	Yes

Table 3 continued

References	Technologies	Data type	Algorithm	Accuracy	Mea.
[116]	LiDAR, IMU	Point cloud, velocity, accelerometer	The sliding-window-based FG formulation	An acceptable accuracy in most scenarios	Yes
[115]	LiDAR, IMU, wheel odometer, camera, GNSS receiver	Point cloud, monocular image, speed, acceleration, orientation	The sliding window based FG method	0.57 m (RMSE), 2.95 m (maximum error)	Yes
[72]	IMU, LiDAR, license plate recognition camera, RSU	Point clouds, license plate number, lane, image, acceleration, angular velocity, orientation	EKF, the C-V2X fusion positioning method	49 cm (maximum average error), 29.67 cm (average error)	Yes
[40]	3D LiDAR	LiDAR reflectivity information	Single correlation matching, EKF	0.35 m longitudinal and 0.19 m lateral (RMS error)	Yes
[71]	LiDAR	Point cloud	RQE-based point-cloud alignment algorithm with a sliding window filter	< 1.8 m (in the damaged section of the tunnel)	Yes
[69]	3D LiDAR	Point cloud	A LiDAR-based map-aided localization method, EKF	< 0.2 m (RMS error in the lateral and longitudinal directions)	Yes
[108]	Ground penetrating radar	Distances to the mine room walls, RSS	PF	–	No
[132]	Ultrasonic	TDoA	Multilateration algorithm	–	–
[98]	Smartphone, LTE	RSSI	Virtual velocitybased subsequence dynamic time warping method	< 22m (average error)	Yes
[83]	Magnetic sensor, smartphone	Magnetic field sequence data	Subsequence dynamic time warping	< 10m (average error)	Yes
[41]	Smart phone	Magnetic field	Magnetic field based positioning	7.32 m (RMSE)	Yes
[75]	Vision sensors, ultrasonic sensors	Distance, image	The barcode locating and recognition algorithm	0.381m (average error in curved underground tunnel), 0.142m (average error in straight corridor)	Yes
[103]	Parabolic wave equation-based wireless channel simulator	DoA	A split-step parabolic equation method, MUSIC, parabolic wave equation methods, weighted K-nearest neighbour and KNN algorithms	1.4109 m (mean error), 0.5092 m (standard deviation)	Yes
[68]	LED light, V2V, V2I, camera	LED signal, image	The intensity filtering	< 1m	No

Table 3 continued

References	Technologies	Data type	Algorithm	Accuracy	Mea.
[42]	RSU, Impulse Radio-UWB, IMU, wheel speed sensors	RSSI, yaw rate	Bayesian data fusion framework based on PF algorithm	–	No
[133]	NXP JN5148	ToF	ToF-based localization algorithm with the error compensation unit	Using static calibration to compensate the error in distance measurements: 1.26 m (mean error) and 2.65 m (max error). Using dynamic calibration to compensate the error: 1.32 m (mean error) and 4.25 m (max error)	Yes
[92]	MIMOSA radio channel sounder	Doppler shift, ToA	EKF, KF	EKF applied on Doppler shift (EKF DS) method: 0–35.88 m (positioning error), 20.30 m (mean value), 11.04 m (standard deviation) and 23.11 m (RMSE). EKF DS and KF ToA method: 0–38.06 m (positioning error), 21.04 m (average value), 12.66 m (standard deviation) and 24.55 m (RMSE). KF applied on Doppler shift (KF DS) and KF ToA method: 0–51.39 m (positioning error), 30.46 m (mean value), 13.27 m (standard deviation) and 33.22 m (RMSE)	Yes

considered as a distinct signal transmitted synchronously from a virtual transmitter under pure LoS conditions to the receiver. In order to reduce the complexity of the calculation, the estimated long visible paths by KEST algorithm were used for vehicle localization. This type of localization method can also be used for vehicle localization in tunnel scenarios.

- Using multipath mitigation techniques to mitigate multipath effects for multipath propagation scenarios. Many multipath mitigation techniques can be used to reduce error in code delay calculation, such as narrow correlator, high resolution correlator or double delta correlator, early-late slope technique, E1-E2 tracker, strobe correlator etc [136–138].
- By using high-resolution signal processing techniques, such as digital beam fouling or beamforming, to accurately locate where a signal is coming from. This can help the system better distinguish between multiple propagation paths and reduce the impact of multipath effects.
- By using filtering algorithms to reduce the noise and errors introduced by multipath effects, the stability and accuracy of the localization estimation is improved [139]. Moreover, machine learning algorithms can also be used that can learn patterns of multipath effects from historical data and make more accurate signal predictions and localization estimates.
- The use of multi-frequency band signals, especially those with higher frequencies, can reduce the impact of multipath effects [140]. Higher frequency signals have shorter wavelengths, making it easier to penetrate obstacles and reducing reflections and refractions.
- The use of directional antennas or antenna arrays can reduce the effects of multipath effects because they can locate the signal source more precisely. In addition, antenna designs can be optimized for specific applications to reduce signal attenuation.
- Environmental map information (e.g., buildings, obstacles, and terrain, etc.) can be used to improve multipath propagation models. By combining environmental information with signal propagation models, multipath effects can be predicted more accurately.

NLoS transmission

NLoS propagation is a common problem, where the signal may experience multiple reflections and scattering within the tunnel, resulting in an unclear signal path. This can lead to inaccurate distance measurements and angle estimates, affecting positioning accuracy. To minimize the affect of NLoS propagation on the location precision, there are many measures that we can take.

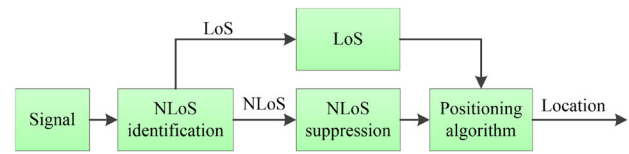


Fig. 14 The framework of vehicle positioning based on NLoS identification and suppression techniques

- Under NLoS propagation conditions, NLoS identification and error elimination are effective methods to improve localization accuracy. Advanced NLoS identification and suppression algorithms are widely used to minimize the impact of NLoS propagation on localization accuracy. The basic framework of this type of localization method is shown in Fig. 14. The received signals may be from LoS conditions or NLoS conditions. The simplest method is to perform NLoS identification first, and then abandon the measurement points in NLoS conditions and directly use the measurement points in LoS conditions for ranging and positioning [141]. However, this method will lead to the loss of useful information. In order to improve the positioning accuracy, NLoS error suppression is required after NLoS identification. Some algorithms can be used to discriminate between NLoS propagation and LoS propagation, such as residual analysis judgment method [142], statistical feature class identification method [143], machine learning methods [144–149], geometric relationship class [150], etc. Then, various NLoS error suppression algorithms, including weighted class, filter class, LoS reconfiguration, neural network algorithms, path tracing algorithm, optimization algorithm with constraints, etc., are employed to alleviate the adverse effects of NLoS propagation on positioning accuracy [151].
- We can incorporate the map information in the tunnel, especially about buildings, road sign and obstacles, that can help the system better estimate signal propagation paths and shadow areas.
- Moreover, the database of pre-collected signal fingerprints can also be utilized to help the system perform better position matching in NLoS situations.

Positioning algorithm

By improving the quality of sensor data, optimizing the performance of algorithms, and utilizing communication and cooperative positioning, the reliability, accuracy, and real-time of vehicle positioning in tunnel scenarios can be improved.

- Cooperative positioning using V2X communication, where vehicles can share each other's positioning information to enhance the accurate positioning of the vehicles inside the tunnel. Currently there are fewer applications of cooperative sensing techniques in tunnel scenarios, the methods of cooperative sensing information fusion need further development.
- Machine learning and DL algorithms are applied to train algorithms to learn environment-specific patterns within tunnels from massive data. We can use these algorithms to enhance the robustness of the localization algorithms and adapt to changes in different tunnel conditions.
- Further development of advanced signal processing algorithms, especially in the face of multipath effects and signal fading in tunnels, in order to enhance the ability of localization system to process signals.
- Adopting an incremental map update strategy to incorporate information on new, changed or repaired tunnels into the localization system in a timely manner to improve the accuracy and usefulness of the maps.
- Using edge computing techniques to shift part of the computational tasks from the vehicle to edge devices inside the tunnel to reduce the computational burden inside the vehicle and improve real-time performance and energy efficiency.
- Combining different positioning modalities such as inertial navigation, visual navigation and GNSS to build a hybrid navigation system to provide more reliable vehicle positioning in different tunnel scenarios.

Limited environmental features

There is usually a lack of distinctive landmarks and environmental features in tunnels, which poses a challenge for feature matching based localization methods. Because lighting conditions are usually poor in tunnels, therefore conventional visual localization methods may not be applicable. To enhance positioning accuracy, many measures can be implemented to increase the information of environmental features within the tunnel.

- Selecting cameras with high sensitivity increases the image quality in low light conditions, which improves the visual system's perception of the environment inside the tunnel. Image quality can be improved by installing additional lighting, such as LED lights, inside the tunnel. These lights provide sufficient illumination to enable the camera to capture better images. Using image post-processing techniques such as denoising, image alignment and feature tracking to enhance the quality and stability of the image data.
- It is also possible that infrared cameras can capture thermal radiation and provide image information even in

completely dark tunnels. This is useful for improving the performance of visual localization in low or no light conditions.

- By using High Dynamic Range (HDR) camera technology, the exposure can be automatically adjusted between high and low light scenes to capture a wider range of brightness and provide more detailed image information.

Cross-tunnel switching

If a vehicle is traveling from one tunnel to another, the wireless positioning system needs to be able to switch smoothly and maintain accurate positioning information, which may require dealing with signal differences in different environments. In addition, when a vehicle enters a tunnel from outside the tunnel, the accuracy of the positioning system will be influenced due to the different signal propagation environments inside and outside the tunnel.

Future work

Currently, there is extensive research on vehicle positioning technology, and many researchers have made significant contributions to tunnel positioning technology in terms of positioning accuracy, system deployment difficulty, and method applicability. However, most methods are limited to theoretical simulations under certain assumptions and conditions, with limited application in real tunnel environments. Few positioning methods have been widely applied in practical settings. Therefore, tunnel positioning technology remains a hot topic in terms of positioning and navigation, requiring further exploration, especially in the context of vehicle localization in tunnel environments.

To address the complex and dynamic environment within tunnels, further research can delve into multi-sensor fusion for vehicle positioning methods. This involves exploring effective integration of the measured data from different sensors, such as LiDAR, cameras, and INS, etc., to improve the accuracy and robustness of vehicle localization. Additionally, research on optimizing sensor layout and configuration to adapt to the specific conditions of tunnel environments is essential.

Introducing deep learning into vehicle positioning systems is crucial for learning and adapting to different complexities of tunnel environments, such as straight and curved tunnels, and effectively handling complex data in tunnel scenarios.

Developing real-time map updating algorithms that promptly reflect changes in tunnel structures and enable real-time map updates is essential. Researching more intelligent and adaptable environmental modeling methods is crucial to better capture and understand the dynamic environment within tun-

nels, achieving three-dimensional environmental modeling in tunnel scenarios. Investigating how to achieve real-time environmental modeling and map updates to respond to changes in tunnel structures and dynamic conditions during vehicle travel is crucial for enhancing the real-time capability and adaptability of vehicle positioning systems.

Researching efficient V2X communication technologies in tunnel environments to facilitate information exchange between vehicles and infrastructures is crucial. Developing more robust V2X communication protocols to ensure reliable data transmission in the confined and obstructed environments of tunnels. Promoting the integration of vehicle positioning systems with V2V communication technologies facilitates collaborative positioning among vehicles. Subsequently, combining vehicle positioning data with real-time traffic management systems to optimize the flow of vehicles within tunnels is essential. Exploring collaborative optimization methods enables vehicles to intelligently coordinate, improving overall traffic efficiency.

VLC offers a novel approach to vehicle positioning in tunnels. Lighting is required in the tunnels, so utilizing existing LED lights for positioning requires no additional energy consumption and can save energy. And visible light communication is not affected by electromagnetic interference, especially suitable for tunnels in such a closed environment, avoiding the interference problems that may be encountered with traditional radio signals. Therefore, utilizing LED lights, which are indispensable in tunnels, for vehicle positioning is a promising positioning strategy. Future research could explore advanced VLC algorithms and their integration with other positioning systems. For example, we can combine VLC with radio frequency systems to leverage the strengths of both technologies and mitigate their individual weaknesses.

Conclusion

In this paper, we presented a comprehensive survey of the key topics for vehicle localization in tunnel environments. It is mainly considered from the aspects of applications, positioning technologies, positioning methods, challenges, as well as future prospects. Many applications of vehicle positioning system were explored. Currently, the application of vehicle positioning in automatic driving systems is perceived as not yet mature. The positioning technologies in tunnels are not as sophisticated as those in unobstructed environments, which need to be further researched in order to improve positioning accuracy. By reviewing various positioning methods, such as the range-based localization, fingerprinting, multi-sensor fusion localization, CP, etc., it can be confirmed that the current trend in positioning method is towards multi-sensor fusion localization method and V2X-based CP approach.

Moreover, we also highlighted a number of challenges affiliated with tunnel localization and provided some solutions that can help in tackling these challenges. We also analyzed some research topics about vehicle positioning techniques in tunnels, including multi-sensor fusion, positioning method based on deep learning, real-time map updating and environmental modeling in tunnels, V2V-based positioning, and VLC positioning, etc. We hope that this article is beneficial to promote the development of more accurate and robust vehicle localization system for the tunnel environments.

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Declarations

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